



Accumulator Design – ENGG4200

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Abstract

Following the Formula SAE Australasia (FSAE-A) competition in 2024 it was determined that one of the development paths for 2025 was going to be a new Accumulator, which is the Energy Storage System of an Electric FSAE car.

The changes to the Accumulator would be a change in style of cells, from cylindrical cells to lithium polymer cells, referred to as pouch cells. These are more energy dense, so it was believed that for the same voltage (453.6 V) there could be a significant decrease in weight. Following competition, it was also determined to reduce the capacity of the Accumulator as in the Endurance event there was left over charge, meaning that there was more capacity than required.

The change in cells alone promised an approximate 7 kg decrease in weight. This project was only to design the mechanical system, not to manufacture or commission the design.

The goals for the design were to lose at least 10 kg, minimise the footprint of the design, and maintain serviceability. Without the full manufacture of the system these goals cannot be stated as completed, however they are trending in the correct direction.

The final design of the Accumulator offers a weight decrease of 17 kg maximum, down to minimum 53 kg from 70 kg. This would also correspond to a decrease to 5.6 kWh capacity.

The volume of the design decreased approximately 32% due to the design of the cell mounting and Container.

No comment can be made on the serviceability until the new one is fully manufactured with wiring and all. This remains a goal of the design and implementation.

The design will now be passed onto succeeding engineers who will complete testing, manufacture, and commissioning of the design. A series of [videos and a presentation](#) to aid handover of the project will occur shortly after the submission of this report.

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1. Introduction

Following the completion of the FSAE-A Competition in 2024, NU Racing's Leaders discussed the development path of the succeeding team. One of the development paths was the design of a new Accumulator with the purpose of losing weight.

The Accumulator is the Energy Storage system of an electric FSAE Vehicle. It stores battery cells that power the Tractive System, which is the electric motor and motor controller.

In previous years, NU Racing have used cylindrical cells in the Accumulator. The current cells have been used for over three years. As they approach the end of their lifetime the team had to decide whether to replace the current cells or purchase a different style of cell and change the design of the Accumulator.

A different style of cell was selected by 2025 Chief Engineer, Alec Chapman. These were Lithium-Ion Polymer Battery, otherwise known as pouch cells or LiPo cells. This was since pouch cells are more energy dense than cylindrical cells and therefore can offer the same energy at a lower weight.

Due to this decision a new Accumulator needed to be designed. NU 24's Accumulator was set to be reused by the 2025 team to reduce the manufacturing load, with changes to ensure rules compliance. Therefore, the design would be applicable to the 2026 team. The author would design the Accumulator whilst the 2025 engineers prepared NU 24's Accumulator for competition, then at the end of the first semester the design would be given to the 2025 team to finalise and begin testing and manufacture.

2. Scope and Goals

The scope of the author's ENGG4200 project were as follows:

- Design of:
 - Container
 - Segment Packaging
 - Top Plate
 - Mounting
 - Lid
 - Maintenance Plugs
- Completion of the Structural Equivalency Spreadsheet (SES)
- Handover of the project design at its completion

Items explicitly not in the scope included:

- Cooling
- Manufacturing and Commissioning
- Physical Testing

The three main goals of the Accumulator Design are shown in Table 1.

Table 1. Accumulator Goals

Goals	Measurable
Minimum 10kg weight reduction	Weight [kg]
Decreasing of Accumulator footprint	Base Size [m ²]
Maintaining serviceability	Decommissioning Time [s]

3. Constraints

The constraints were:

- The BMS, IMD, IRs, and Current Sensor must be reused
- Because the BMS was being reused the number of cells needed to remain the same (108 total cells)
- The pack would have a 108s0p configuration (108 cells in series, zero in parallel)
 - This would result in the same voltage as NU 24's Accumulator but a reduction to 5.6 kWh in capacity

4. Assumptions

The assumptions were:

- The Accumulator would not require active cooling (cooling achieved with a fluid such as air or water)

5. Accumulator System Introduction

The Accumulator System is made up of three subsystems: the Segments, the Container, and the Accumulator Electronics and High Current Path.

5.1. Segments

The Segments are made up of the cells, Segment Housing, Busbars, and a Segment Printed Circuit Board (PCB) that records temperature and voltage of each cell.

Inside of the housing the cells are arranged in series, parallel, or a combination of the two to provide a pack voltage and capacity.

Each segment requires a PCB which records temperature and voltage of each cell. In NU 24 this PCB is called the CANaMon.

5.2. Container

The Container contains the Top Plate, Segments, and High Current Path. Its role is to protect the cells and High Current Path in case of a crash and insulate the Tractive System Voltage from the rest of the Chassis.

The Container is made of the Lid, Internal and External Walls, Floor, and Mounts. These all have minimum thicknesses dictated by the FSAE Rules [1]. If the container is made from an electrically conductive material, it must be insulated from the poles of each segment.

5.3. Accumulator Electronics and High Current Path

Inside of the Container the Accumulator Electronics are stored. This includes the Battery Management System (BMS), Insulation Monitoring Device (IMD), Precharge Board, Low Voltage Distribution Board (LVD), Isolation Relays (IR), Fuse, and the Current Sensor. The BMS, IMD, IR, Fuse, and Current Sensor are all bought parts.



The High Current Path is the culmination of wiring and busbars that carries current from the batteries to the Motor and Motor Controller. This begins with the busbars between each cell, then to the busbar/HV wire that leads to the IRs. These relays separate the High Voltage from the rest of the car. From the IRs the path is continued through HV Wires to the HV Connector at the Lid before it is connected to the rest of the car.

6. Introduction to Pouch Cell Mounting

Mounting pouch cells in FSAE is very different from how pouch cells are usually mounted in industry. This is due to the requirements of the FSAE Rules [1] that required that a Segment must be able to withstand the following accelerations:

- 20g in the vertical direction
- 40g in the lateral direction
- 40g in the longitudinal direction

Due to this the cells need to be rigidly mounted to the Segment Housing to avoid failure and non-compliance.

There are generally two methods of mounting pouch cells in FSAE, friction loading through compression plates, or gluing the cells to an external frame. These are shown in Figure 1.

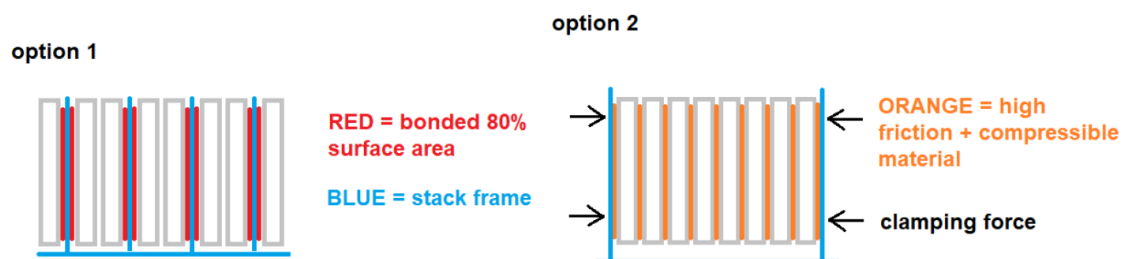


Figure 1. Pouch Cell Attachment [2]

It should be noted that friction fitting is becoming less common in Formula Student in Europe as it requires physical testing (seen in EV 5.5.10)

7. Initial Concepts and Designs

As the first concepts were being developed the main two focus' was on the mounting of the pouch cells into segments, and the size of the segments.

7.1. Segments

7.1.1. Number of Segments

The size of the segments was dictated by the total number of cells allowed by the BMS, 108. Deciding how many cells to place in one segment was as simple as dividing the number of cells by the proposed number of segments (to gain a whole number). The feasible number of segments is shown in Table 2.

Table 2. Number of Cells in a Segment

Number of Segments	Number of Cells in a Segment	Number of internal walls	Weight of Cells in one Segment [kg]	Total Cell Weight [kg]	Segment Voltage [V]
4	27	3	7.344	29.376	113.4
6	18	5	4.896	29.376	75.6
9	12	8	3.264	29.376	50.4

Initial sketches and dimensions of the Container with these segment sizes were created and shown in Table 3.

Table 3. Initial Container Area Calculations

Number of Segments	Minimum Longitudinal Length [m]	Minimum Lateral Width [m]	Minimum Area [m ²]
4	0.26	0.35	0.09
6	0.17	0.55	0.10
9	0.12	0.89	0.10

As the current width and length of NU 24's Accumulator is 525 mm and 520 mm the four-segment design was selected as it allowed for a footprint that would fit inside of the current Accumulator's footprint. This would cause minimal changes to the Accumulator position and

mounting in the Chassis and minimise the number of large changes required, reducing the workload required to reposition the Chassis and Drivetrain in NU 26. The Drivetrain would be especially hard to move to accommodate Accumulator changes without an increase in Chassis size.

As the thickness of the internal walls of the Accumulator are dictated to be minimum 2.3 mm by the FSAE Rules [1], it also seemed that by minimising the number of required walls the overall Container weight would reduce. This would also reduce the number of PCBs and Maintenance Plugs required.

7.1.2. Cell Attachment

As discussed in Introduction to Pouch Cell Mounting there are typically two methods of mounting pouch cells, by friction via compression or with glue and a frame.

The positives and negatives of each mounting solution are shown in Table 4.

Table 4. Positives and Negatives of each Cell Mounting Solution

Glue	Positives	Negatives
	Allows consistent bonding	Requires bonding of every cell
	Bond strength only needs to be credible for one cell	Requires a frame to be made for every cell
	Mounting to a Segment Housing simplified	Cell cannot be removed from a frame - frame design change means new cells will need to be purchased
	Allow for heat transfer from cells to frame and Segment Housing	
Compression	Positives	Negatives
	Allows for changes in mounting design	High coefficient of friction required to keep the cells stationary
	A faulty cell can be changed easily (without needing a new frame made)	High friction non-flammable materials are hard to source
		Rules in Formula Student in Europe require mandatory testing for compression designs
		Creates a large thermal mass of cells

Initially a compression style mount was the main design point. This was majorly due to the fact that singular cells could be replaced without the need to replace others. An initial design is shown in Figure 2.

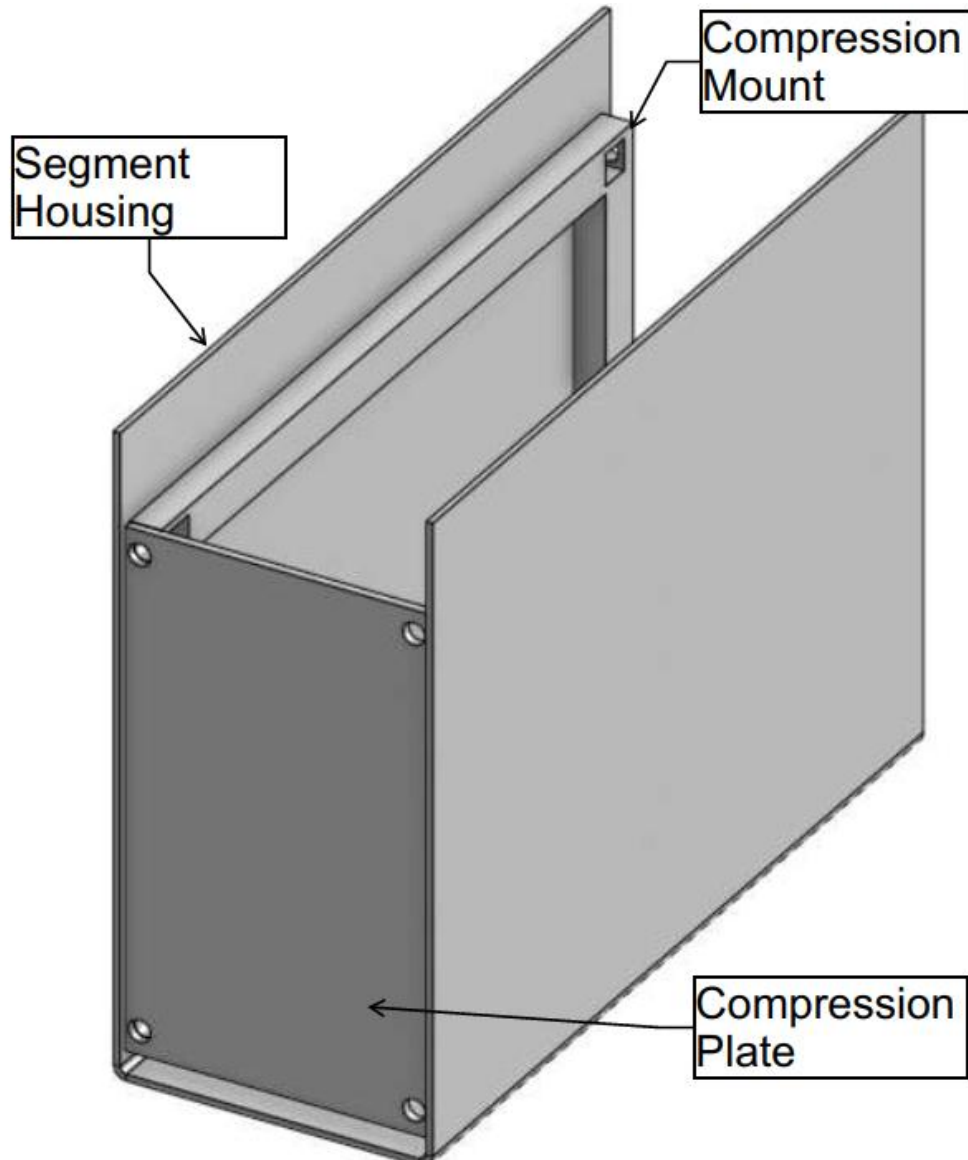


Figure 2. Initial Segment Housing Design

This design was made from Aluminium and had the Compression Mount that would be welded to the Segment Housing. The cells would be placed in the Housing then a Compression Plate would be used on either side of the Housing to hold the cells. Once tightened, the Compression Plate would apply a normal force to the cells.

For the cells to remain in the housing whilst under the 20g and 40g loads as dictated by the FSAE Rules the required friction force and coefficient was calculated. For the cells to remain stationary the friction force had to be greater than the force generated by the acceleration loads.

For the vertical 20g load, with the cell weight of 7.371 kg the vertical force was calculated in Appendix A as 1446 N. To overcome this assuming a material with a friction coefficient of 0.8 could be found a normal force of 1807 N would need to be applied to the face of the cells. This corresponded to a 138 kPa pressure over the cells surface.

Knowing that the 40g longitudinal acceleration would require double the force the author reached out to the cell manufacturer, Melasta, to enquire whether they would advise this loading or not. The response received did not instil confidence and the author was not put in contact with an engineer, instead receiving additional advice from a salesperson. According to the salesperson the maximum force that the cells could withstand was 13 kN, however they immediately then went on to say they recommend a physical distance between the cells to allow for heat dissipation and allowing the cells to expand under charge and discharge.

This knowledge combined with the unknown cooling characteristics of the cells, and the fact that compression designs were undergoing more strict scrutiny in other Formula Student competitions encourage the author to move towards a glue design.

7.1.3. High Current Path

The electrical configuration of the total pack was 108s0p which meant the configuration of the Segments needed to only be in series. For the four-segment design this meant that there was a 27s0p configuration.

There are many methods to connect the cells in series. An example is shown in Figure 3.

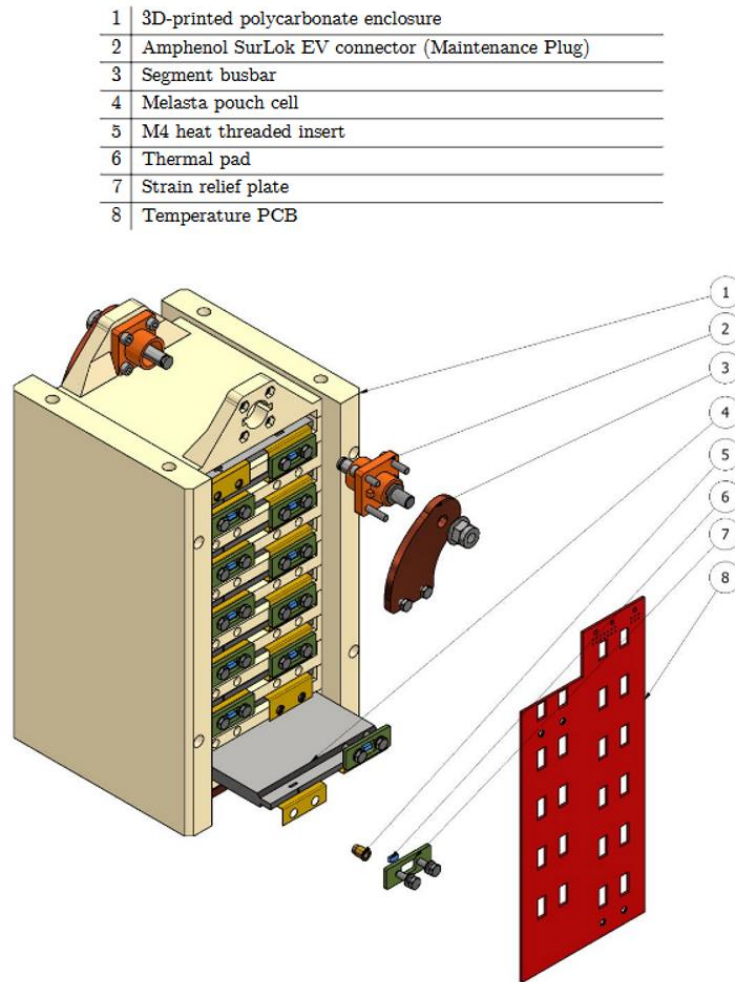


Figure 3. High Current Path Design Example #1 [3]

To connect pouch cells in series one terminal from a cell is attached to the next cell, however they cannot have the same polarity. They should only ever be connected to the opposing polarity (i.e. connecting positive to negative only) to combine cells in series. This is shown in Figure 4.

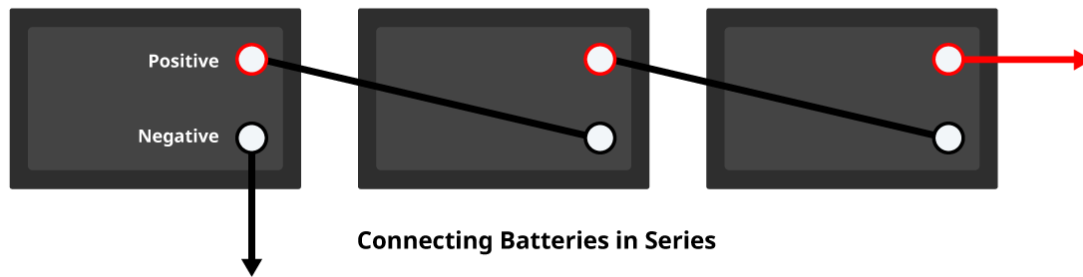


Figure 4. Connecting Cells in Series [4]

To connect the tabs of the current cells in series the tabs of the cell must be clamped or bolted together such that there is a good termination between the two connecting faces.

Initially to connect the cell tabs of the cells together a Clamping Tab design was created. The intent was to have two simple waterjet cut parts instead of many busbars as it was thought that clamping a large number of busbars would become tedious and troublesome to work on.

The Clamping Tab design is shown in Figure 5.

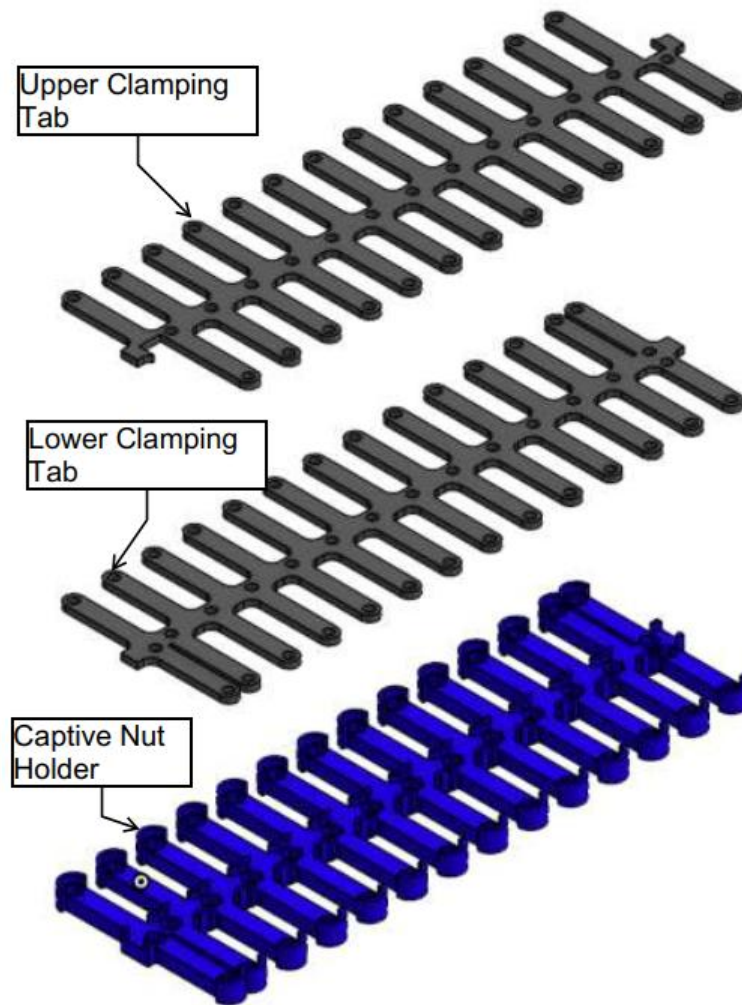


Figure 5. Clamping Tab Design

The cell tabs would be folded over each leg of the Clamping Tab and then the Upper and Lower Clamping Tabs would be bolted together.

The main issue with this design was the large risk of a short circuit if either of the Clamping Tabs were made from a conductive material as they both touched every cell connection. A non-conductive and non-flammable material was sought after. However, as no compressible materials are allowed in the High Current Path (EV.6.4.1. c. [1]) this design was disregarded.

The other large issue with this design was the difficulty in measuring the voltage from each cell and the temperature of each cell. As nothing could be measured from a 'busbar' or easily

from the terminals this would cause the Mechatronic Department great difficulty when creating the required Segment PCB.

7.2. Top/Side Plate

In the design process, the Top/Side Plate was the last thing to be designed and followed the shape of the container dictated by the Container and the Segments.

The initial desire was to have the Top Plate components placed on top of the Segments to decrease the footprint of the Container. However, due to the height of the new Segments this would have made the new Accumulator approximately 100 mm higher than the previous iteration.

Due to the limited area above NU 24's Accumulator and the poor results of the Tilt Test in 2024 [5], conveying that the centre of gravity height of NU 24 was higher than desired, the author elected to place the weight lower in the container and extend its length by 100 mm. This resulted in the Top Plate components to be moved to the rear of the Container.

As one of goals of the Accumulator Design was to maintain serviceability time, the Top Plate components were placed on a new mounting plate that slid in and out of the Container like a Segment. The reasoning for this was to maximise accessibility to the components. By allowing them to be removed from the Container easily, this would hopefully aid maintenance. This assembly was dubbed the Side Plate.

8. Final Design

The Accumulator is made from four Segments, a Side Plate, and the Container. An exploded view of the Accumulator is shown in Figure 6.

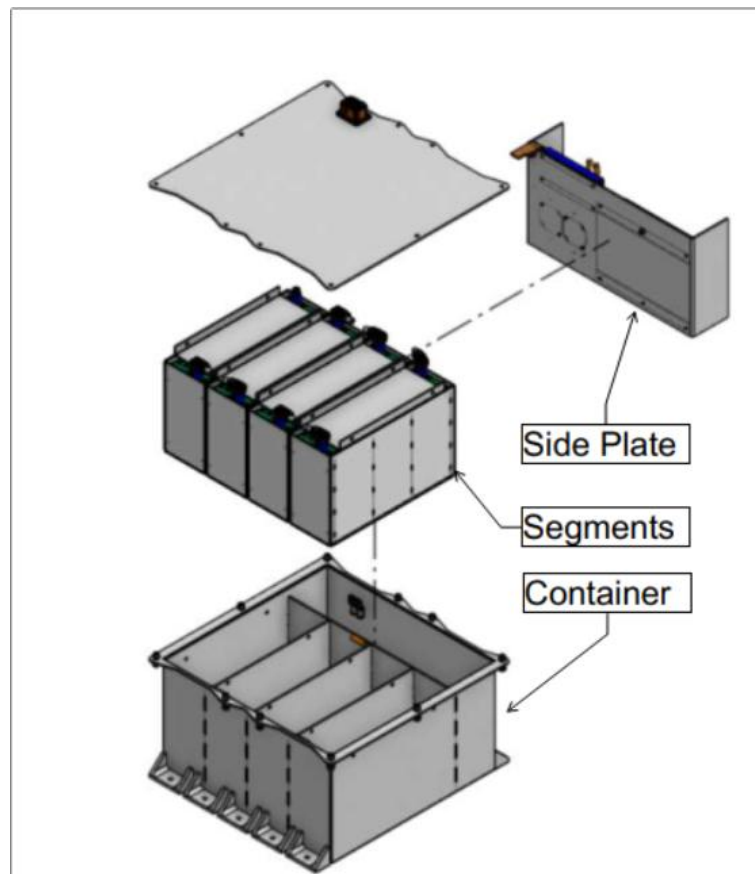


Figure 6. Final Accumulator Design

8.1. Segment Housing

The final Segment Housing design is shown in Figure 7.

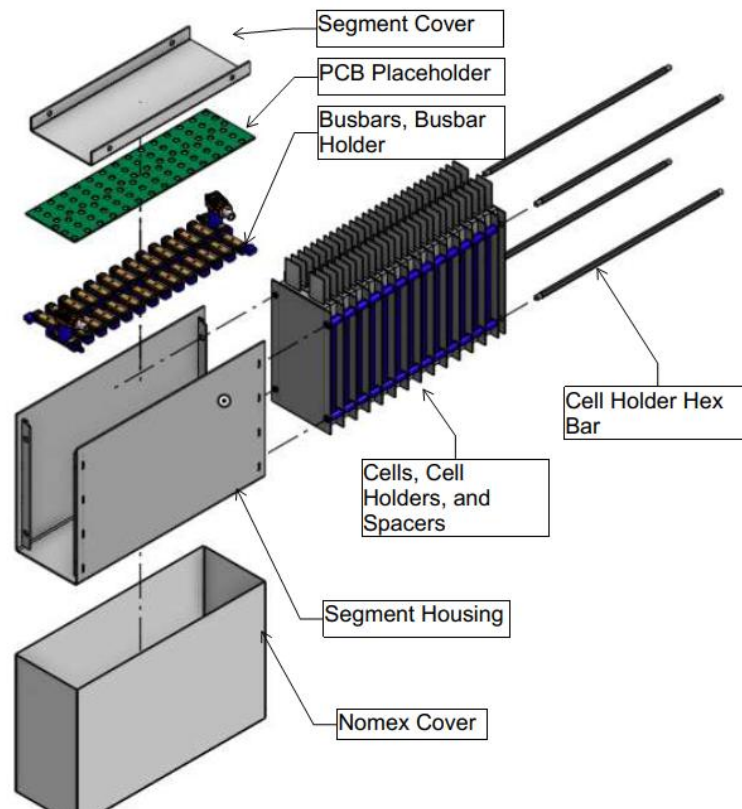


Figure 7. Segment Housing Assembly

The Segment Housing is designed to be made from 5052 Aluminium. It is folded with slots in it to facilitate weldment of the Cell Holder Mounts as shown in Figure 8.

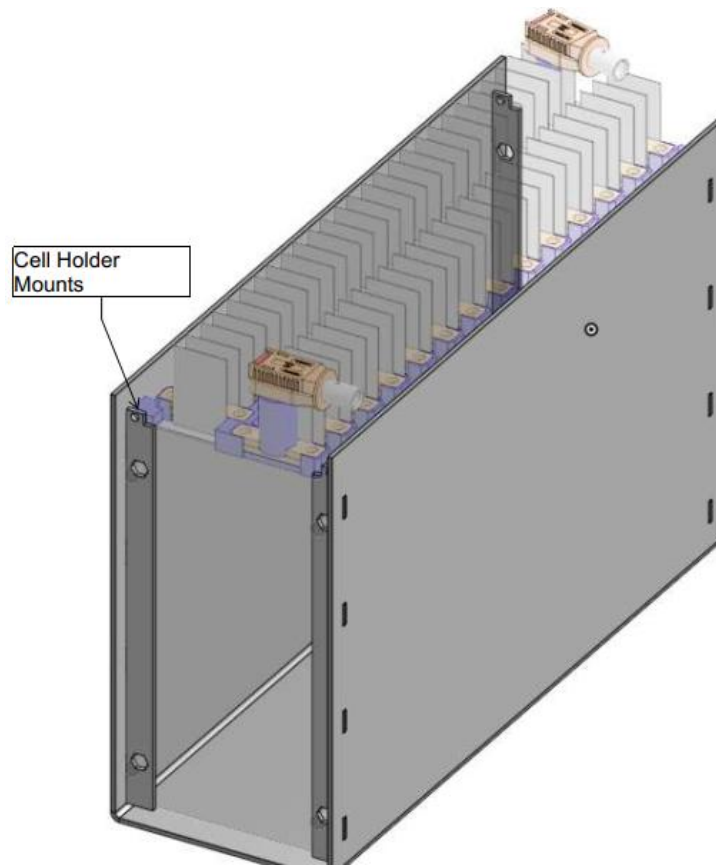


Figure 8. Cell Holder Mounts

The Cell Holder Mounts are designed to carry the acceleration loads of the cells and have hex slots cut into them to allow for the positive locking of the Cell Holder Hex Bar. These are made from $\frac{1}{4}$ " (6.35 mm) 12L14 steel hex bar and are threaded at each end to allow for a washer and glenloch nut to be attached. The Cell Holder Hex Bar feeds through each of the Cell Holders, which are shown in Figure 9, which also have hex slots in them.

The Segment Housing is made from 5052 Aluminium to reduce manufacturing time during the first iteration of the design. 6061-T6 Aluminium is a stronger material, however due to its T6 heat treatment it cannot be easily bent. It cracks and fractures during the process. Using 5052 allows for the Housing to be single piece and bent, with additional Cell Holder Mounts welded in after the fact. 5052 is analytically strong enough, as discussed in Calculations and FEA, at a thickness of 2 mm, however it is possible that 6061-T6 could be utilised in a thinner gauge and

maintain the same strength, whilst being lighter. However, this would require the Segment Housing to become three piece (two side pieces, and a base) and required weldment.

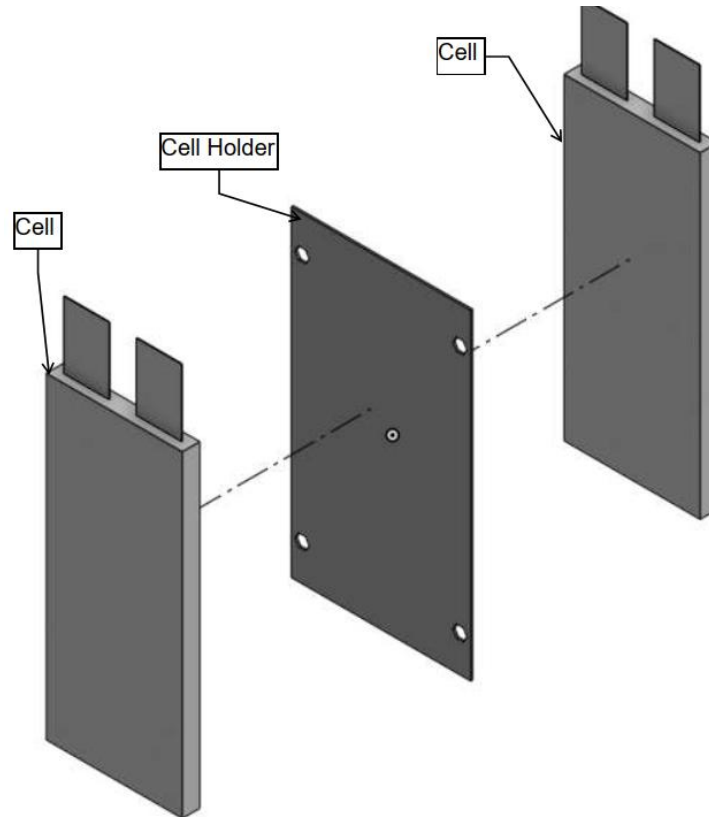


Figure 9. Cell and Cell Holder

The Cells and Cell Holders are attached by gluing. The intended epoxy for this is [834B Black Flame Retardant Epoxy](#) from MG Chemicals. This was used in NU 24's Accumulator with no issues. Each Cell Holder, which is made from 1.02 mm 6061 – T6, is glued to two cells. This halves the number of Cell Holders required (saving 2.8 kg), at the cost of tying cell dependencies together. If one cell needs to be replaced it means that both cells glued to Cell Holder will need to be replaced.

To hold each cell group apart 3D Printed Spacers are used and shown in Figure 10. The main reason for using these is that the thickness epoxy on each cell will not be the same. By making these parts 3D printed (at least for the manufacture of the first full Segment) means that the thickness of the Spacers can be altered to suit the specific group. This is vital, as under

acceleration loads the Cell Holders will try to slide along the Cell Holder Hex Bar, and these spacers stop the cell slamming into each other and limits the relative movement between the cell tabs and the body of the cell. Over time if there was relative movement this may cause fatigue failure of the tabs and cause a short circuit of the entire Accumulator voltage.

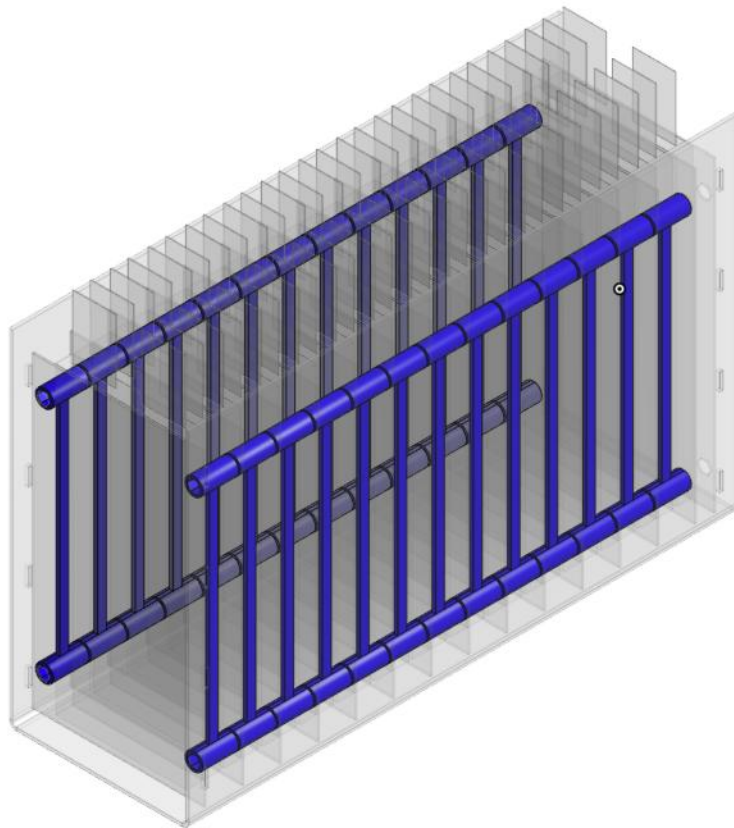


Figure 10. 3D Printed Spacers

The Spacers have a hex pattern cutout to allow for them to index on the Cell Holder Hex Bar. One Spacer is used for the upper and lower Cell Holder Hex Bar on each side respectively to make it easier to position and hold during assembly.

It is suggested that at minimum a 0.5 mm gap is maintained between the groups of cells, at least during the manufacture of the first Segment, as it is unknown how much the cells will swell under charge and discharge. If any concerns are raised, it may be worth attempting to get a swell estimate from the manufacturers (which will be unlikely based on previous communication).

The assembly of the Cell Holders, Cells, Spacers, and Cell Holder Hex Bar is shown in Figure 11.

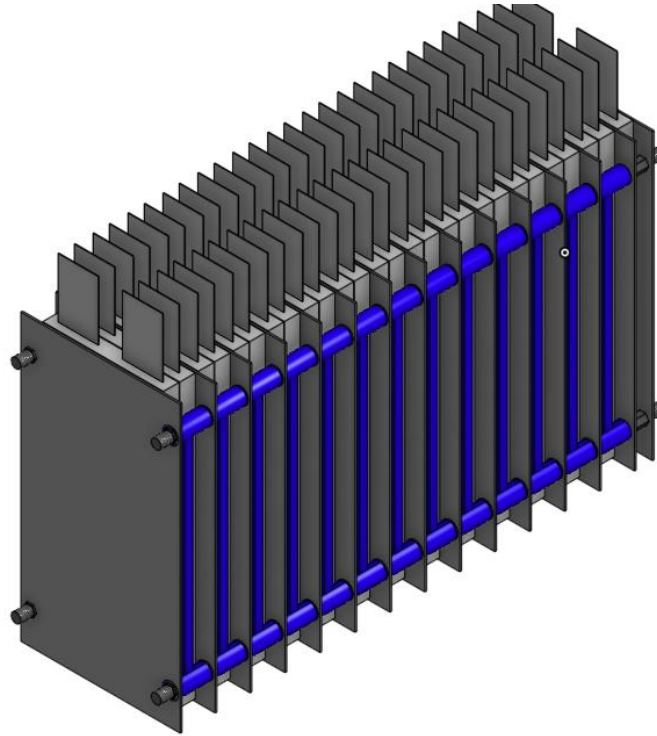


Figure 11. Assembly of the Cells, Cell Holders, Spacers, and Cell Holder Hex Bar

The Busbars and Busbar Holder are shown in Figure 12.

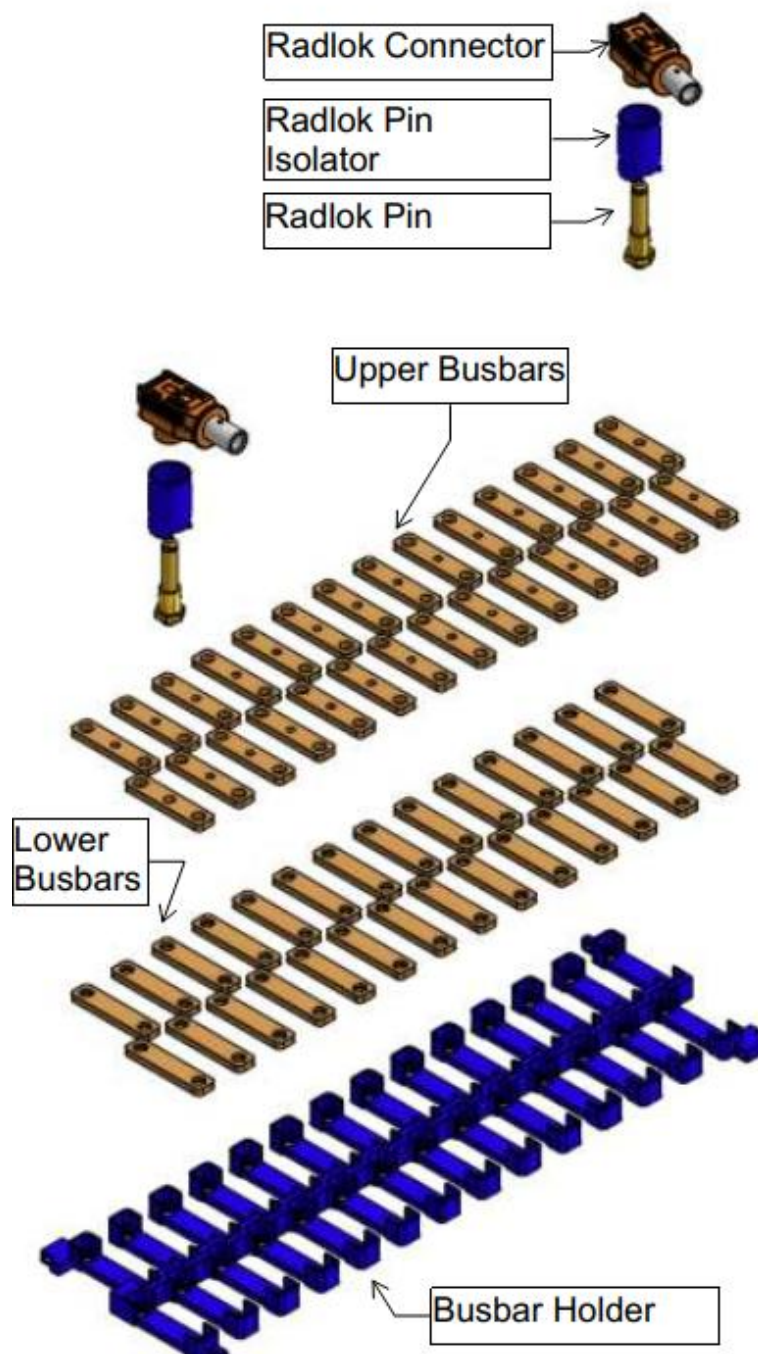


Figure 12. Busbars and Busbar Holder

The Clamping Tab design described in High Current Path, as discussed had a glaring issue which was measuring the voltage and temperature of each cell. To evolve from that initial design the same shape was kept by the Busbar Holder, which is 3D printed. Now using live busbars instead of a non-conductive material means that the voltage can be read from each Upper Busbar, the way it is currently done in NU 24, and the temperature can also be read from

these Busbars. The Upper Busbars have threaded centre holes which are intended to be bolted to the Segment PCB (however this can, and is encouraged, to be changed to suit requirements by the Mechatronic Engineer designing the PCB. Both the Upper and Lower Busbars are made from copper.

The Busbar holder has two functions, to hold the Upper and Lower Busbars, but to also hold the captive nuts underneath the Lower Busbars. The nuts are sat in the holder prior to the Lower Busbars being sat in there as shown in Figure 13. This was done to reduce handling of the smaller parts in a High Voltage Situation, especially when the Accumulator is open.

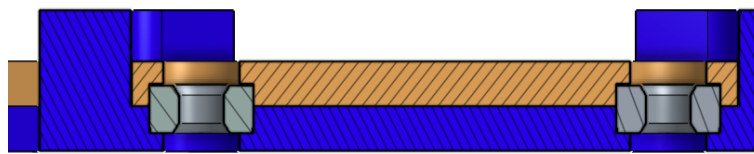


Figure 13. Section View of the Lower Busbar and Captive Nuts

To positively lock the bolted connection holding the Busbars together, the nut is positively locked (has its ability to rotate) locked by the Lower Busbars, which have a machined hex shape slot on their underside, and the intention is that the bolt will be positively locked by the CANaMon replacement PCB. The Lower Busbar machined slot and NU 24's CANaMon PCB is shown in Figure 14 and Figure 15.



Figure 14. Lower Busbar Captive

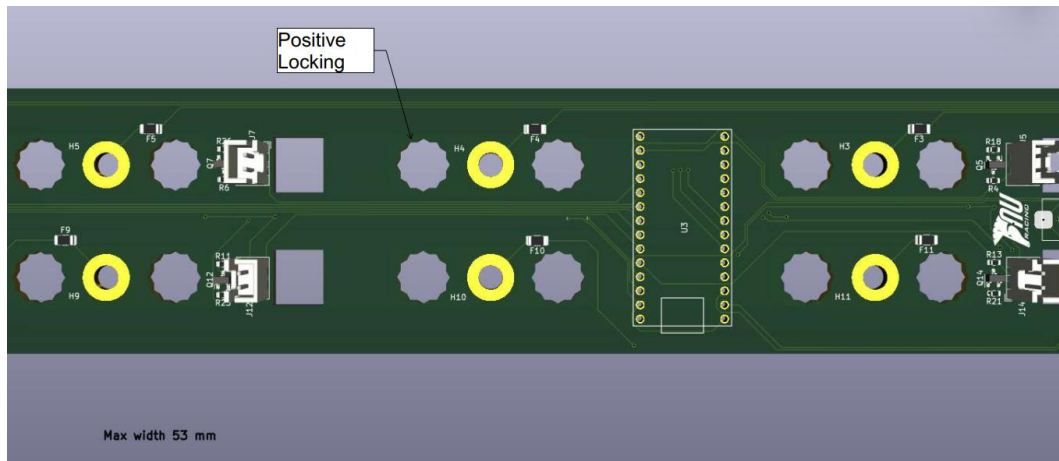


Figure 15. CANaMon Positive Locking

Manufacturing the Lower Busbars will be incredibly tedious as there is 112 to manufacture, so it is suggested that these are outsourced to an external company, like PCBWay or equivalent. Prior to this they will need to be 3D printed and checked for any issues before a bulk order is placed. Please note that they have already been 3D printed by the author, however this should still occur again prior to purchase to ensure that there are no issues (such as a change in the CAD that was forgotten about).

The Busbar Holder is mounted to the top of the Cell Holder Mounts to stop movement of the Busbars and PCB relative to the top of the cells. This connection is shown in Figure 16

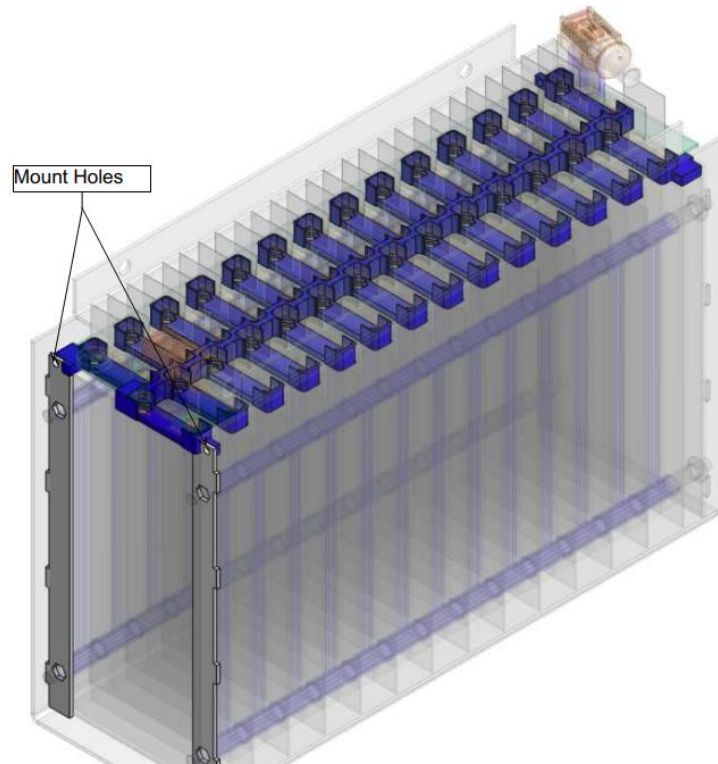


Figure 16. Busbar Holder Mounts

The Segment Cover is made from 5052 Aluminium and had the role of containing the Segments under vertical accelerations. This works similarly to the Lid of NU 24's Accumulator, as it places a compressive load on the Segment Housings. Each Cover is bolted by an M6 bolt to the Internal Walls of the Container. The Segment Housing extends higher than the top of the top of the CANaMon equivalent, meaning that the Segment Cover only interacts with the walls of the Segment Housing. This is shown in Figure 17.

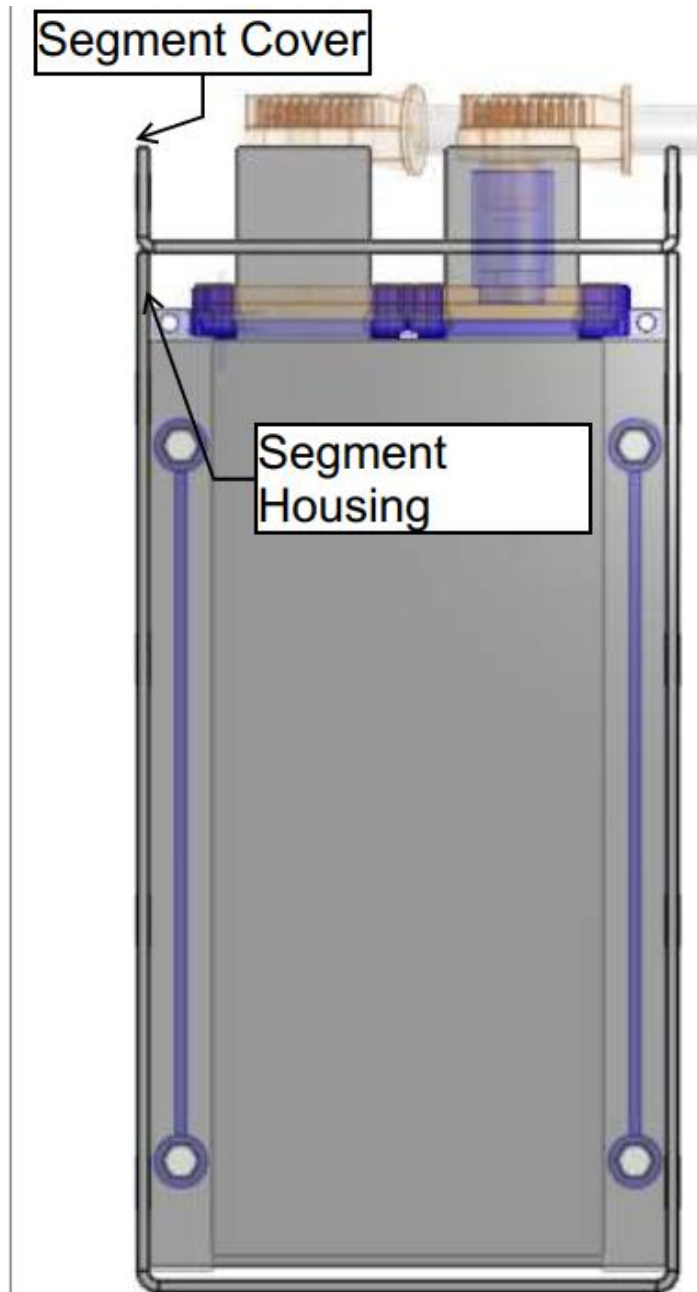


Figure 17. Interaction between the Segment Cover and Segment Housing

To connect an individual Segment to other segments and to the IRs, Radlok Connectors are used. Radlok Pins, shown in Figure 12, are threaded into the Radlok Busbars (which are the first and last Upper Busbars of a Segment). These allow for a Radlok Connector to be used to connect all the Segments to create the total Accumulator Voltage.

8.1.1. Assembly Instructions

To assemble a Segment the following steps should be followed (assuming everything is manufactured and ready to install).

To assemble the Cell Holders to the Segment Housing:

1. Ensure heat shrink is covering the tabs of every cell and cannot fall off easily – if this is done the cells can be installed without HV gloves
2. The first Cell Holder is placed in the Segment Housing (touching either of the most outward Cell Holder Mounts)
3. All four Cell Holder Hex Bar rods are fed into the corresponding Cell Holder Mount past the inner face of the Cell Holder Mount
4. A Spacer is placed onto the upper and lower Cell Holder Hex Bar (repeated for both left and right sides)
5. The next Cell Holder is then inserted into the Housing, and the Hex Bar is fed through it
6. Steps 3-4 are repeated for every remaining Cell Holder
7. Once the final Cell Holder and Spacer are in position a M6 washer and glenloch nut are fastened to each of the Cell Holder Hex Bar ends

To connect the cells in series:

1. Insert all the M5 nuts (normal nuts NOT glenloch nuts) into the Busbar Holder
2. Insert all the Lower Busbars into the Busbar Holder above the M5 nuts
3. To decrease the hazard of a short circuit the cells need to be connected in series in a fashion to minimise the voltage of every connection
 - a. To achieve this complete Steps 4-7 in the following order:

- i. For every connection on the right-hand side (Maximum Segment Voltage: Nominal Cell Voltage)
- ii. For Every second connection on the left-hand side (Maximum Segment Voltage: $3 \times$ Nominal Cell Voltage)
- iii. Complete the rest of the left-hand side connections going from one end of the Segment Housing to the other. The order of this is shown in Figure 18. (Maximum Segment Voltage: Final Connection is $\frac{1}{2}$ of total Segment Voltage)

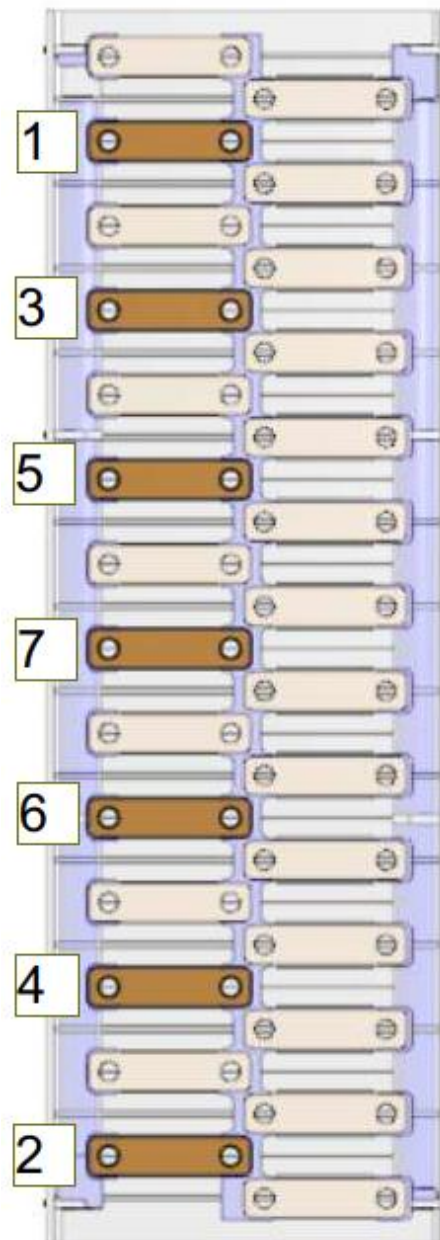


Figure 18. Order of Final Cell Connections during Busbar Assembly

4. Fold the cell tabs from both cells on either side of the Busbar over the Lower Busbar.
5. Place the Upper Busbar on top of the folded cell tabs
6. Insert and fasten the two M5 Busbar bolts.
7. Test that each cell tab is tightly held by the clamping force of the Busbars
8. Fasten the Radlok Pins to the Radlok Busbars.

9. Place the Segment PCB on top of the Busbars and fasten (dependent on the final design of the PCB)

8.1.2. Correlation to the Design Goals

8.1.2.1. Maintaining Serviceability

The Segments are the most complicated sub-system of any Accumulator design and often taken the longest time to physically assemble. Due to this extreme care had to be taken in the design to not excessively increase the serviceability times.

Due to the nature of pouch cells and how they must be connected in series, there was always going to be an inherent loss of assembly time for the Segments, due to the time it takes to fold every cell tab over a busbar before clamping them together.

To aid this, the Busbar Holder design was thoroughly thought out. This design allows for the M5 nuts and Lower Busbars to be inserted into the Busbar Holder prior to being inserted into the Segment Housing. This achieves two things. It decreases the likelihood of a small M5 nut being dropped into the Segment Housing, which would cause risk of shorting and require disassembly to remove and allows the small assembly to be constructed outside of a high voltage environment. If this was not achieved, more time would have been taken to ensure the safe insertion of the nuts, which would increase total serviceability time.

The assembly of the Cell Holders to the Segment Housing also had the potential to increase serviceability time. Using the Cell Holder Hex Bar to mount the Cell Holders to the Segment Housings meant that only four rods and eight washers and nuts were required to mount every cell to the Housing. It does mean that 28 Spacers are required however as these are parts that do not need to be fastened, instead just sliding onto the Hex Bar, the impact of this part count does not greatly affect the assembly time. As the Spacers can be placed in with each Cell Holder

mount, with nearly 100 mm thickness between the sides of the Segment Housing, are able to be easily placed in position, as discussed in later in Prototype.

8.1.2.2. Weight Reduction

The approximate weight of the new Segment design is approximately 10.3 kg. This means that the over four Segments the total weight is 41.2 kg. 2024's Accumulator had a Segment weight of 6.4 kg across nine Segments for a total weight of 57.6 kg. This meant that the Segment design had hit its weight loss goal by losing 16.4 kg in the Segments alone. This does include a theoretical weight for the Segment PCB but until this is designed and manufactured its weight is unknown. Similarly, the actual weight for a Segment will not be known until one is manufactured.

8.1.2.3. Footprint Reduction

The Segments take up a volume of 6.5 L. With four Segments this is a total volume of 26 L.

The NU 24 Accumulator Segments took up 3.2 L for a total volume of 29 L.

This is an approximate volume reduction of 3 L.

8.2. Side Plate

The Side Plate replaces NU 24's Top Plate and houses the electrical components of the Accumulator. The Side Plate is shown in Figure 19.

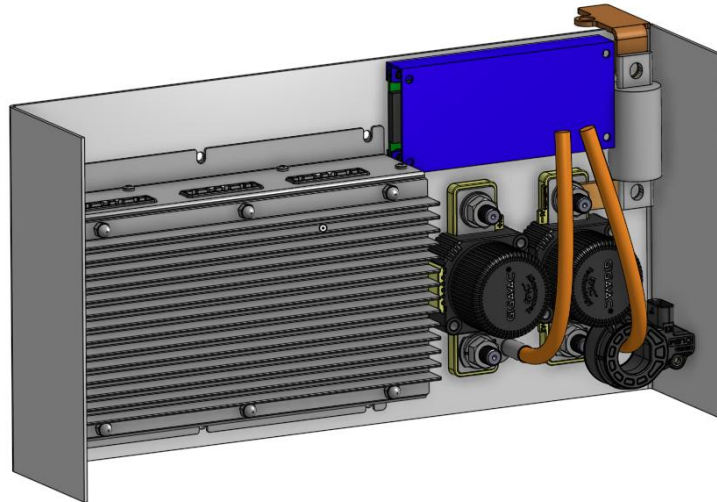


Figure 19. Side Plate

The Side Plate houses the BMS, IRs, IMD, Fuse, Current Sensor, and Thermal Sticker Busbar. The Side Plate components are shown in Figure 20.

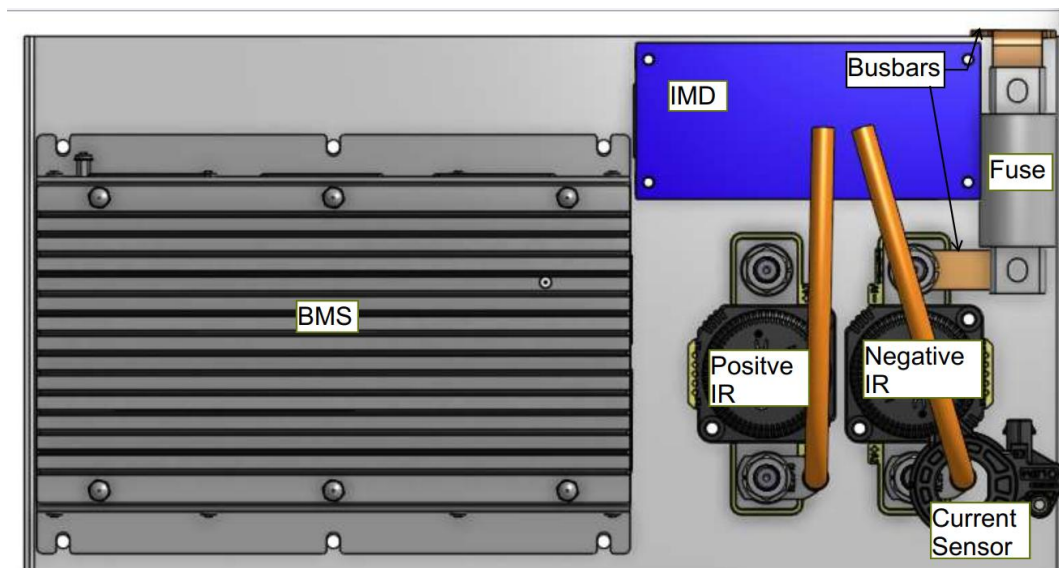


Figure 20. Side Plate Components

Note that the rightmost IR is the negative IR and such has the Tractive System Fuse connected before it. The left IR is the Positive IR.

The Side Plate components are all bolted to a single folded sheet metal part. This is also intended to be made from 2 mm thick 5052 aluminium sheet. This was selected as the components on the Side Plate are not held to the same requirements as the Segment Housings,

meaning it does not need to be designed to withstand the 20g and 40g acceleration loads. Instead, it simply needs to hold the components steadily during normal driving conditions. It is intended that due to the complex array of mounting holes, that all the holes are laser/waterjet cut. The mounting holes are shown in Figure 21.

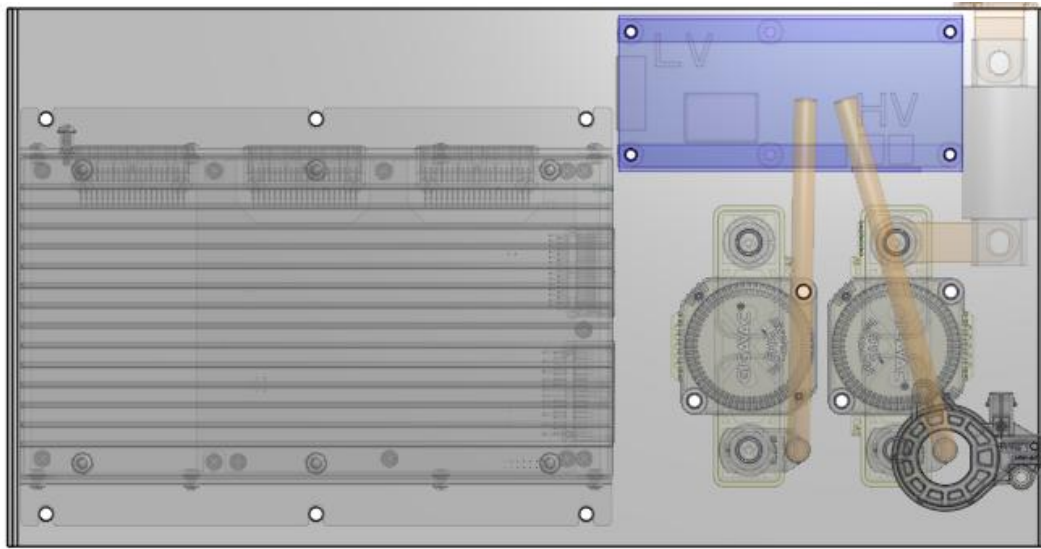


Figure 21. Side Plate Mounting Holes

Each component is bolted with the head of the bolt sitting behind the Side Plate, shown in Figure 22. To ensure the Side Plate is insulated from the rest of the Container, the entire plate must be wrapped in Nomex Sheet (front, rear, and both flanges), and the head of the bolts covered in Kapton Tape. This is to limit the number of surfaces that could become live due to an Accumulator failure, such as a short. Due to the bolts sitting behind the Side Plate, it is critical that there is appropriate tolerancing given between the heads of the mounting bolts and the Rear External Wall of the Container. Currently this is set to give 0.2mm, however this should be reviewed prior to manufacture in case of changes to the mounting bolts (the thickness of the head of the bolts will change the distance between the Side Plate and the Rear External Wall).

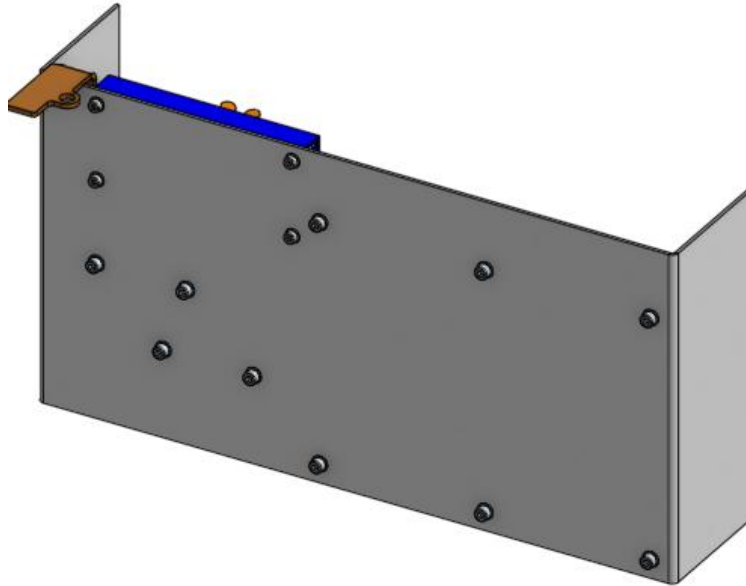


Figure 22. Head of the Fasteners on the Rear of the Side Plate

Either side of the Side Plate has flanges that are intended to keep the Side Plate vertical during its install and removal from the Container. The direction of removal is shown in Figure 23. The flange on the right side of the Side Plate is intended for the mounting of a new Precharge Board. The current Precharge Board is wider than depth of the Side Plate and does not fit. To refit the board there are two possible positions to consider, rather the right flange of the Side Plate or the top of the Side Plate above the BMS. These areas are shown in red in Figure 24. If it cannot fit within these limits, it is suggested that the Side Plate is extended up to the height of the Lid. If this is not possible, mounting the board to the Lid should be explored first instead of extending Side Plate depth to mount it next to the HV Fuse, as this will extend the length of the whole Accumulator.

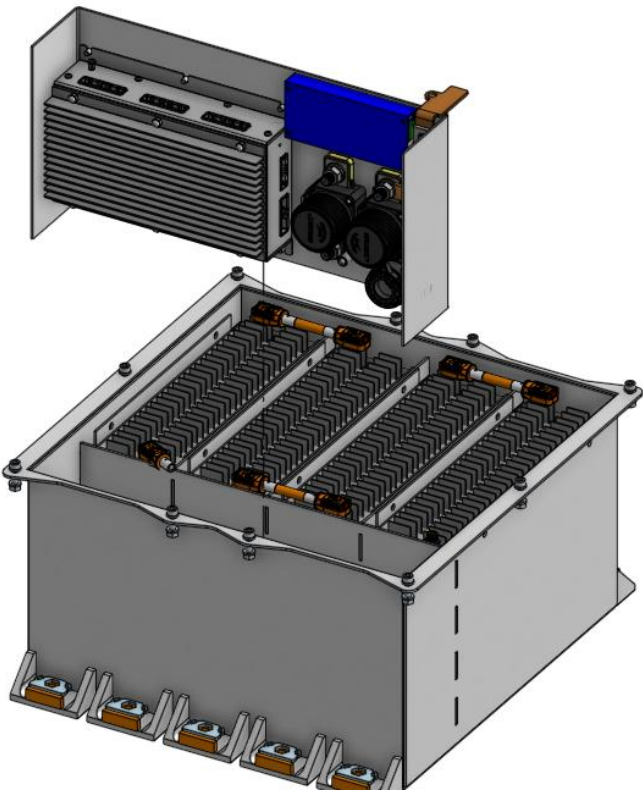


Figure 23. Direction of Removal of the Side Plate Assembly

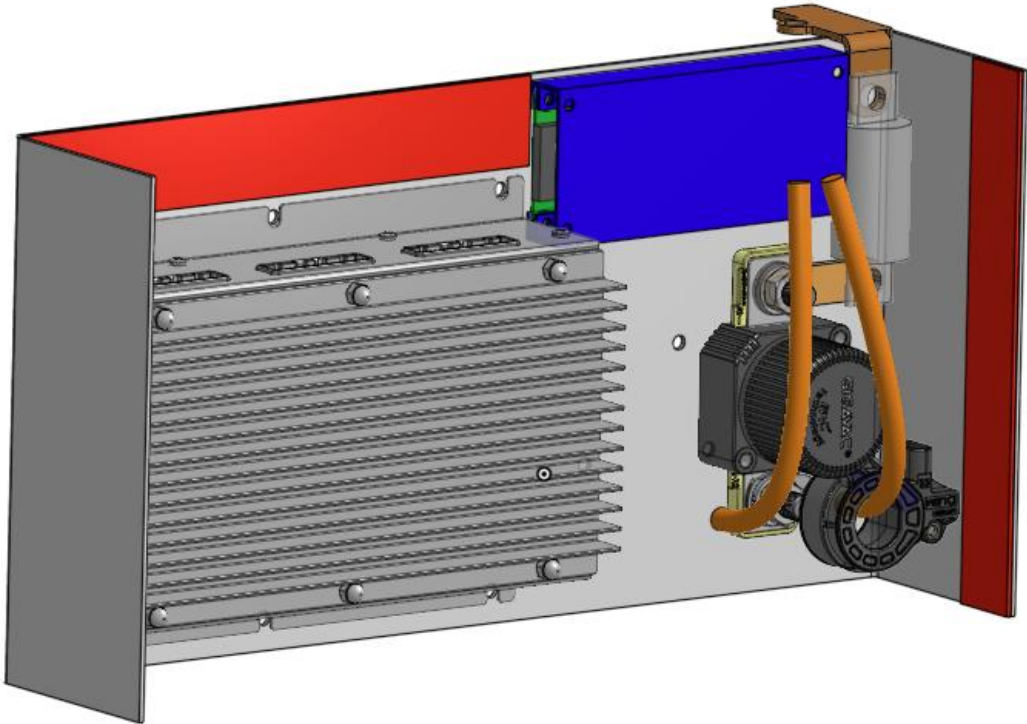


Figure 24. Precharge Board Intended Area

The two HV cables that connect to the IRs then travel to the lid as the output poles of the Accumulator. This is shown in Figure 25.

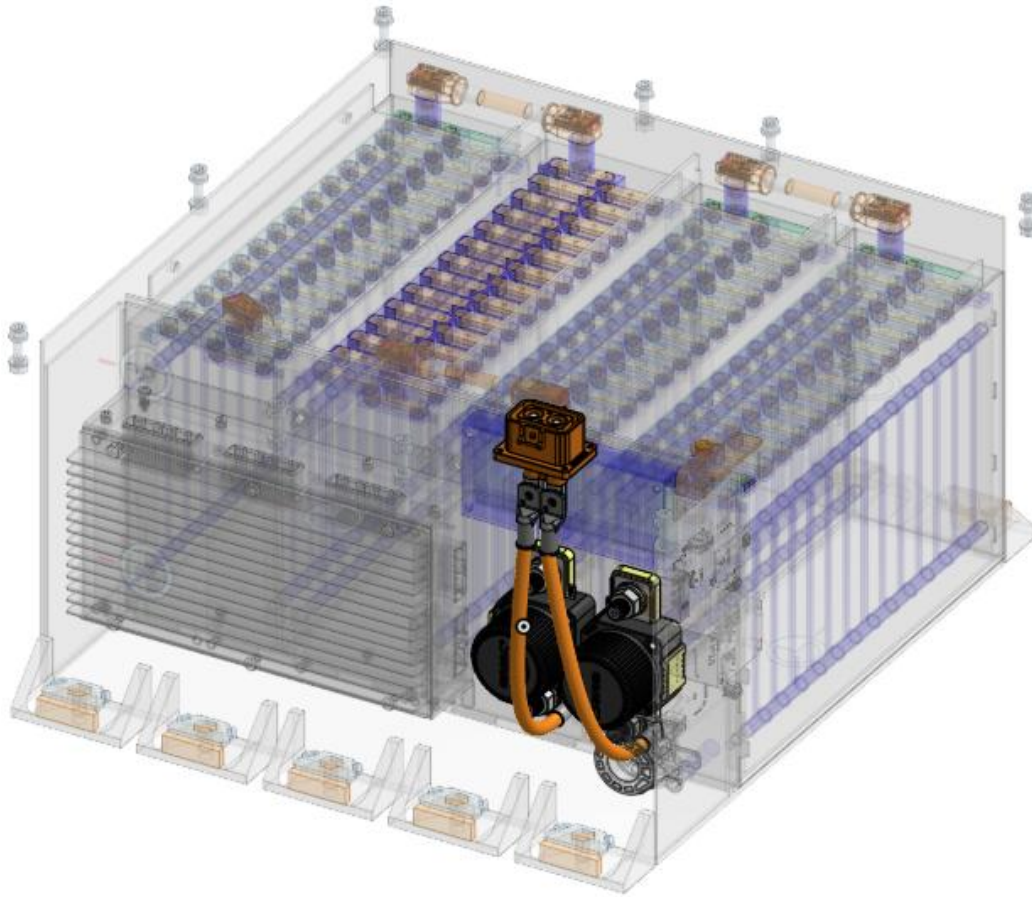


Figure 25. IR Connection to Lid Connector

As the path the 25 mm² wires travel has a tight radius, 90° Cable Lugs are used to assist in making this transition. An example of these is shown in Figure 27. The HV cables have a minimum bending radius which is 5 times the outer diameter. The OD of 25mm² HV Wire is 11.4 mm, and therefore a minimum bend radius of 57 mm. The required bend radius is shown in Figure 26. It should be noted that the sketch shown in Figure 26 only shows the side view of the cables and such only the radius in that plane. The cables are also free to rotate in the front plane as shown in Figure 20. This is also the absolute minimum length of these cables, meaning if they were made this length, it would be near impossible to lift the Lid of the Container. Once these wires are made, they must be larger than this length to allow for removal

of the lid, and with this increase in size the wire will have more slack to bend with, increases its bend radius.

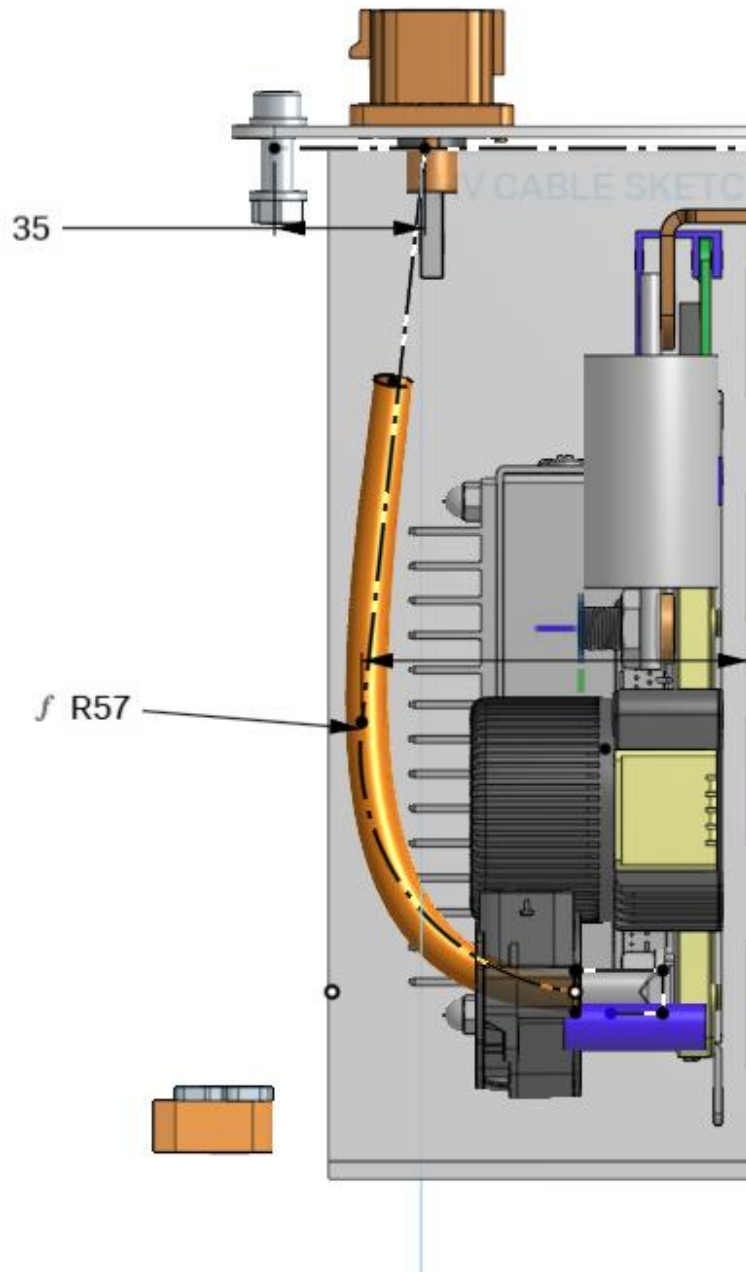


Figure 26. HV Wiring Bend Radius



Figure 27. 90 Degree Cable Lug [6]

NU 24's Accumulator utilises two Radlok Pins and Connectors that are fastened to the Top Plate as means of a quick disconnect for the HV cables that lead to the Lid. This is shown in Figure 28.

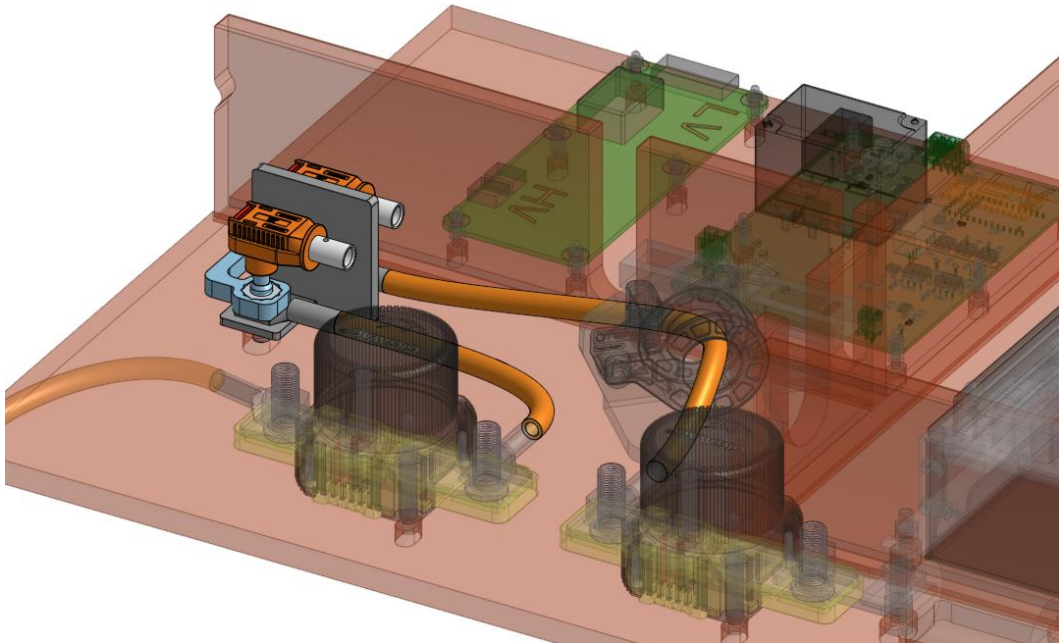


Figure 28. NU 24 Lid HV Quick Disconnects

These Radlok Connectors Currently rely on the Radlok Pins that are Fastened to the Top Plate. To replace the need for excessive parts the author suggests using an Amphenol Surlok Straight Plug and Inline Crimp Termination Receptacle. These are shown in Figure 29.

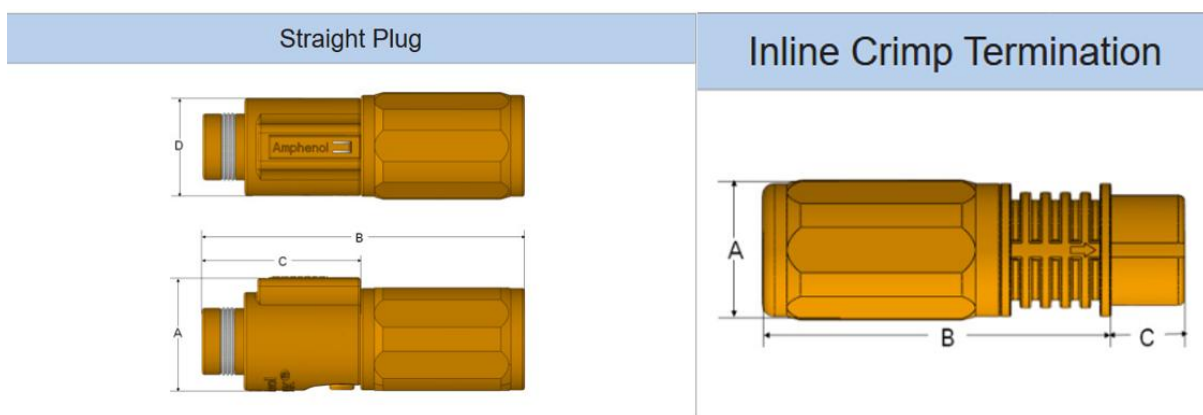


Figure 29. [Surlok Straight Plus and Receptacle](#) [7]

These connections will allow for the Quick Disconnect to become an in-line feature instead of requiring mounting.

The size of the Side Plate is driven by the width of the Segment Housings and thickness of the Internal Walls. Due to this the design space to fit all the components became extremely tight. Of note, the clearance between the IMD and the BMS, and the clearance between the IMD and the Fuse were of major concern. A prototype of the Side Plate was created and is discussed in Prototype. These areas are still of mild concern and call for the Prototype to be fully utilised with the actual BMS, IMD, and Fuse to ensure clearances.

EV.6.1.2 states that with all enclosure in place, contact with any Tractive System connections with a 100 mm long, 6 mm diameter probe must not be possible. In the Side Plate the areas that will need to have covers designed and implemented are the IRs and the Fuse connections (both to the Busbar and to the IR. It is assumed that the Busbars will be insulated with heat shrink.

Mounting of the Current Sensor is incredibly difficult due to the tight configuration of the Side Plate and the limited space where the HV cables are run. To simplify the location of the Current Sensor, as opposed to mounting it to the lid (where there is very little space between the Positive and Negative HV Wire), the author proposed removing one of the mounts of the Orion Current Sensor. The interfering mount is shown in Figure 30.

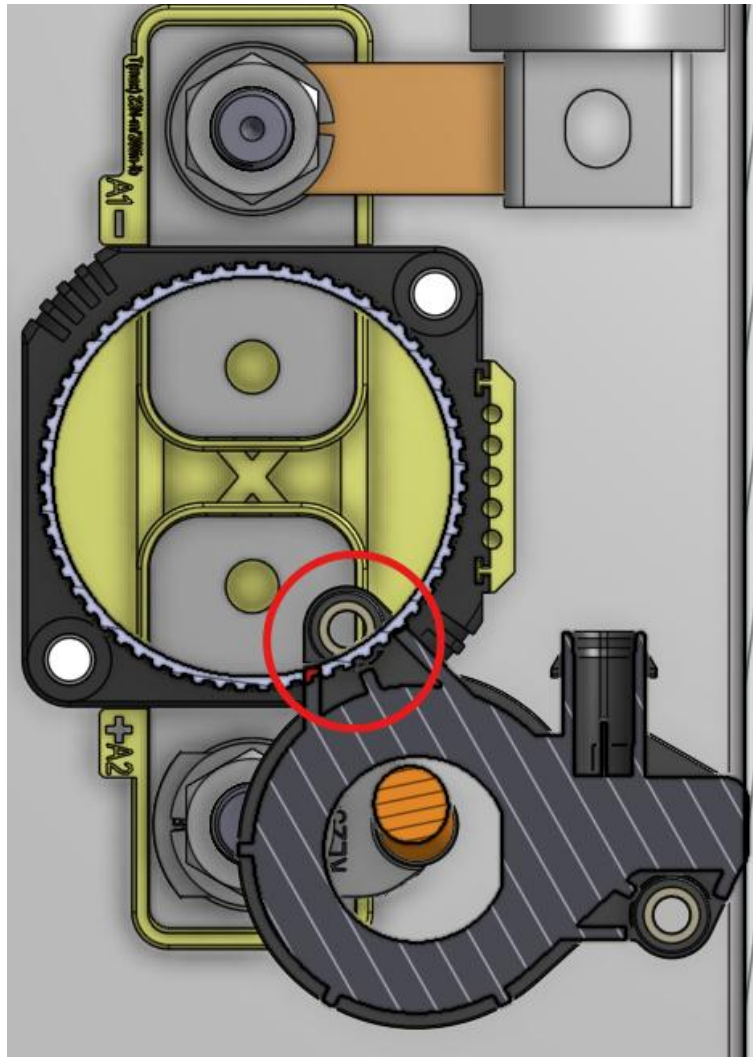


Figure 30. Current Sensor Interference with IR

The reason to remove the mount is due to the required proximity of the Current Sensor to the IR and the orientation of the connector of the Current Sensor. Keeping the connector facing upwards gives the largest clearance for the connector.

8.2.1. Correlation to the Design Goals

8.2.1.1. Maintaining Serviceability

The Side Plate is purely designed with serviceability in mind. Moving the Accumulator components from the Top Plate to the side of the container was always going to be a challenge and introduce difficulties in servicing connectors and mounting of hardware. By allowing the Side Plate to be removed from the Container for service, it reduces access and clearance issues

for connection/removal of connectors and the usage of tools. By allowing this, the time it will take to removal the Side Plate and begin maintenance is assumed to be very similar to NU 24's Top Plate.

8.2.1.1. Weight Reduction

The Side Plate assists largely in the removal of weight of the Accumulator by removing the structural Top Plate of NU 24. As this was a structural member it meant that it had to be nearly 10 mm in thickness. As it covered the nearly the entire footprint of the Accumulator, it was nearly 3 kg alone. By removing this and replacing it with a single 2 mm thick aluminium sheet metal part, which weighs approximately 700 g, the Side Plate can save a considerable amount of weight.

8.2.1.2. Footprint Reduction

The Side Plate adds to the base area of the Accumulator instead of adding vertical height. The Side Plate takes up approximately 8.4 L of volume compared to NU 24's Top Plate which occupied approximately 16.4 L of volume. This is half nearly half of the volume.

8.1. Container

The Container for the Accumulator is designed as a welded 6061-T6 Aluminium Container. This is made with slots that allow for external weldment, except for one section of weldment for the Segment Internal Walls.

The Container is shown in Figure 31.

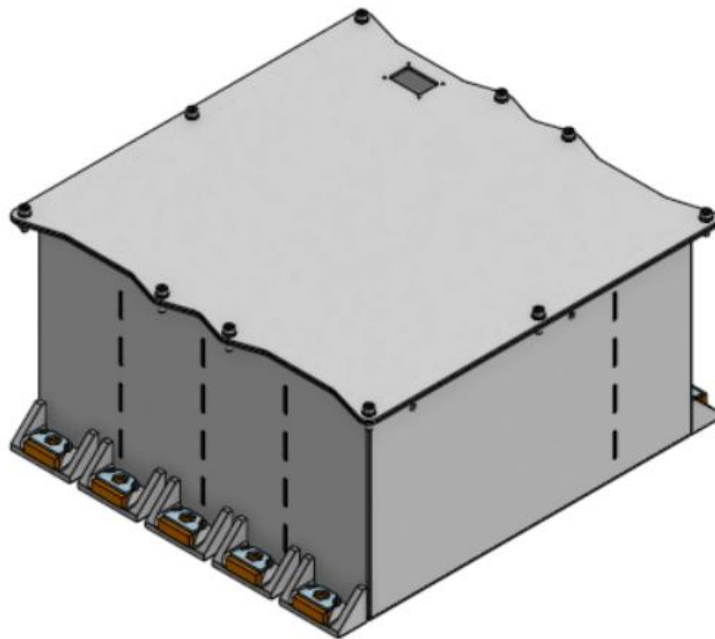


Figure 31. NU 26 Accumulator Container

The Container is made from the Internal Walls, Floor, Lid, External Walls, and Mounts. This is shown in Figure 32.

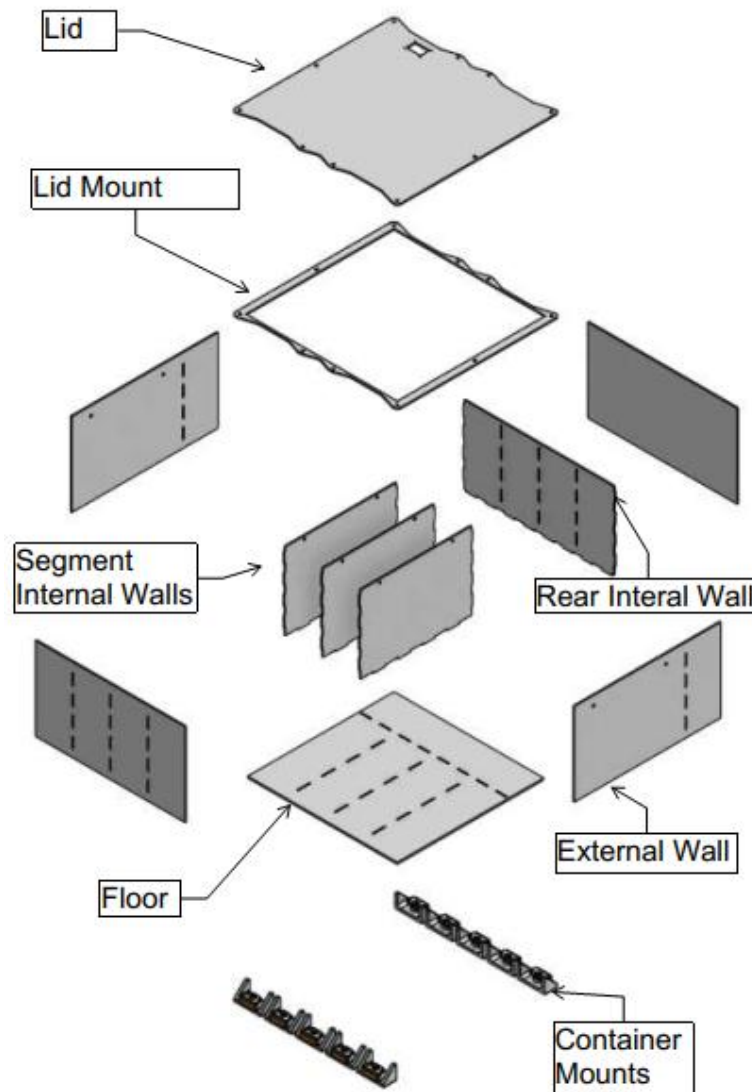


Figure 32. Container Assembly

The Internal Walls, External Walls, Lid, and Lid Mount are all designed from the minimum thickness 6061 – T6 Aluminium available from sponsor Calm Aluminium whilst being compliant with the minimum thickness required for the walls and lid (which is 2.3 mm [1]), 2.54 mm. The Floor is similarly made from the minimum complaint thickness for the floor and lid (which is 3.2 mm [1]), 4.06 mm.

To aid in the weldment and assembly for weldment of the container weld slots have been reutilised as they were in 2024. However, the 2024 rules state that all welds must be greater than 25 mm and the weld/space ratio must be equal to or higher that 1:1 (F.10.2.5 [1]).

Currently in NU 24 there are welds that are 10 mm long both lateral external walls (behind both Fan Manifolds) as shown in Figure 33.

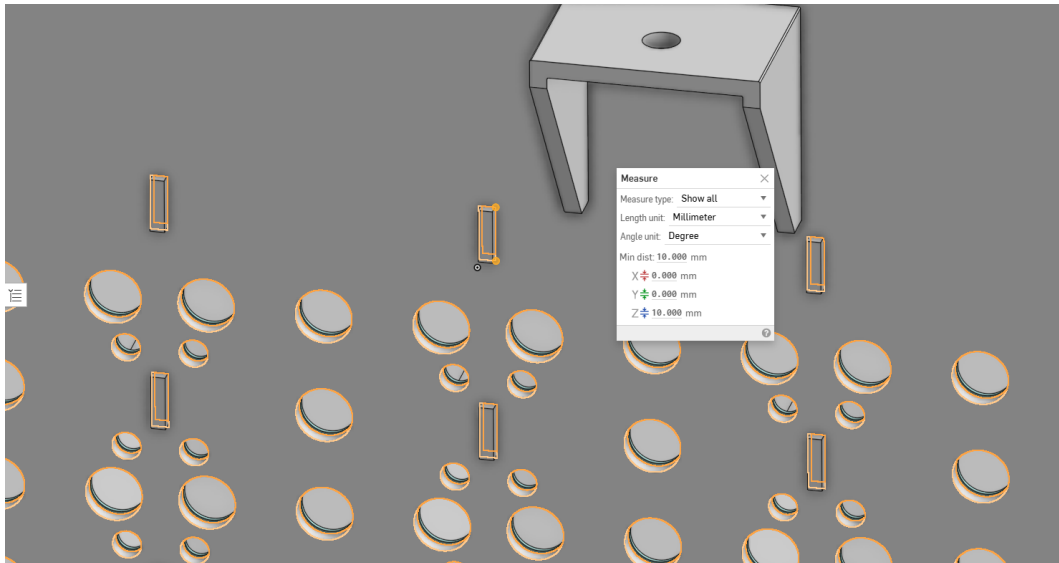


Figure 33. NU 24 Container Welds

These are not compliant and have been updated for the NU 26 Container.

There are no holes in the container for airflow. This is due to the belief that the new cells will not require active cooling (such as fans or water cooling).

This may require that a pressure relief valve must be installed, according to the 2024 Technical Inspection Document, however in the 2025 Rules it states that “Segments sealed in Accumulator Containers must have a path to a pressure relief valve”. It is not known if the current design would be considered a sealed Container, and a rules enquiry should be submitted as such.

The Container is split into two main sections. One houses the Segments and the other houses the Side Plate. The areas are shown in Figure 34.

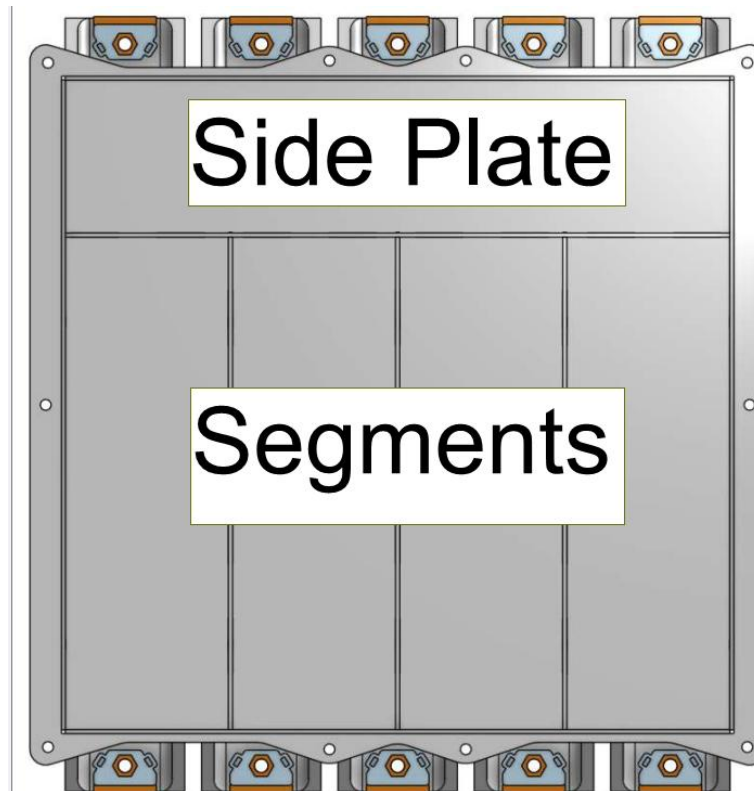


Figure 34. Container Areas

As discussed in Top/Side Plate the side plate was move to the rear of the Container and as such is being treated as another 'Segment' that can be removed prior to maintenance. To fasten the Segments and the Side Plate, bolts extend through respective Internal and External Walls. The mounting holes for the Segment Cover are highlighted in Figure 35.

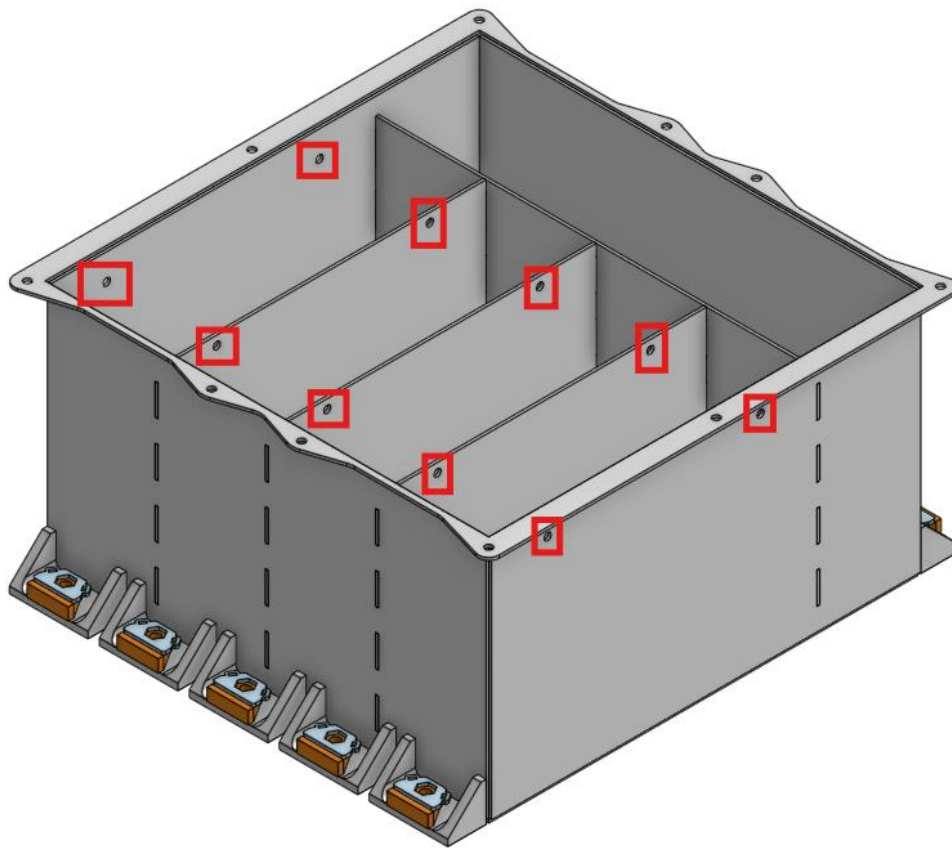


Figure 35. Segment - Container Mounting Holes

For the holes protruding through the External Walls, EV.5.2.2 c. applies which states “Any conductive penetrations, such as mounting hardware, must be protected against puncturing the insulating barrier” [1]. This means that the end of the bolts that feed into the Container need to be covered with an insulative cover. This has not yet been designed. It is suggested a 3D print is used to cover the fastener inside of the Container. Note that this needs to be rigid enough such that it cannot fall off under acceleration loads. Therefore, a friction fit is not suggested, but instead a solution like how the LVD is mounted to the Lid in NU 24’s Accumulator. This would be a 3D printed cover that goes over the bolt inside of the Container, then two screws are fastened through the Container from the outside into the 3D print cover. There are only six of these external fastener attachments in the Container (four for the Segmetns and two for the

Side Plate), so whilst this is tedious, it is not expected to take large chunks of maintenance time.

The Container Mounts are the exact same ones used for NU 24. This is as NU Racing still has the material in the Prototyping Workshop metal racks and its SES compliance. 10 total mounts are still required as the mounting is still designed under Load Based Analysis (F10.5.7 [1]). This states that an Accumulator with weight greater than 40 kg requires 10 attachment points. Corner Based mounting (as described in F.10.5.6 [1]) was not considered an option as creating a solution that needed to be bolted from all eight corners of the Container would require more complex manufacture and increase the time to install/remove the Accumulator from the Chassis.

The Lid, unlike NU 24's Lid, does not take the load from the movement of the cells, and as such is not reinforced. It is simply a flat piece of aluminium with mounting holes for itself and for the Accumulator connectors (such as the LVD and the Accumulator Output Connector).

The Segment connections to the Container are made by the Segment Cover as described in Segment Housing.

8.1.1. Correlation to the Design Goals

8.1.1.1. Maintaining Serviceability

To maintain the serviceability of the Container the same mounts from NU 24 are reused. This ensures that the Accumulator can still be installed and removed from the bottom of the Chassis with 10 bolts in minimal time.

The lid was simplified to allow for easy installation and removal.

8.1.1.2. Weight Reduction

The designed Container has an approximate weight of 8.5 kg measured from Onshape. This compared to the 12.7 kg weight of the NU 24 (also measured in Onshape for comparison)

grants the new design an expected weight reduction of 4.2 kg. This weight is product of the loss of internal walls, eight internal walls to five, a smaller overall Container size, and a simplified lid and mounting solution.

8.1.1.3. Footprint Reduction

The footprint reduction of the NU 24 Container to the new Container is shown in Figure 36, Figure 37, and Figure 38.

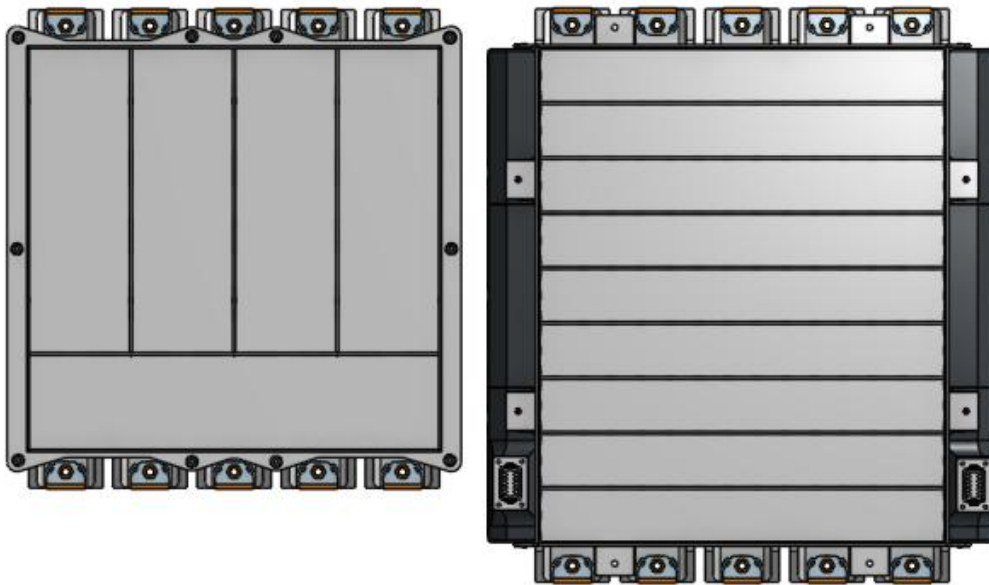


Figure 36. NU 26 Container (Left) vs. NU 24 Container (Right) from a Top View

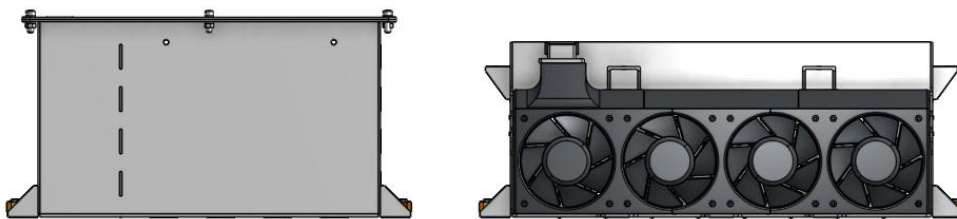


Figure 37. NU 26 Container (Left) vs. NU 24 Container (Right) from a Side View

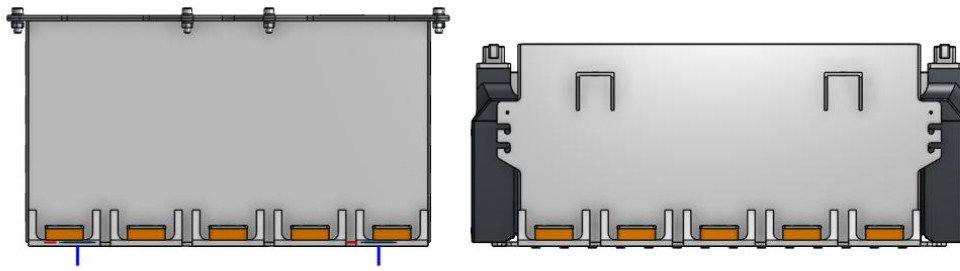


Figure 38. NU 26 Container (Left) vs. NU 24 Container (Right) from a Rear View

The volume reduction to 51 L for the new Container from 75 L for the NU 24 Accumulator. This approximately a 32% reduction in volume. This is the total Accumulator volume reduction.

8.2. Insulation

One of the most vital parts of an Accumulator design is its insulation. This prevents the Container from becoming electrically live and suppresses fire in the case of an Accumulator failure.

Insulation in previous Accumulator designs most frequently took the form of FR-4 and GPO-3. As one of the goals of the new design Accumulator was to decrease weight, it made sense to also optimise the weight of the insulation. One of more commonly used Accumulator insulations in FSAE is Nomex Paper. At thickness above 0.13 mm [Nomex Paper 410](#) recognised as UL94V-0, which is required (F.1.18 [1]), and has a basis weight of 249 g/m² at a thickness of 0.25 mm thickness [8]. This means that lining every internal surface of the Accumulator (at 0.25 mm) would weigh approximately 351 grams. Comparing this to NU 24's insulation, the weight of the lining on just the Container (not the Lid) was 510 grams.

Moving from FR-4 to Nomex means that it will not be required to epoxy the FR-4 to the Container, which in 2024 was a lengthy and expensive process. Instead it is suggested that the Nomex Sheet is attached to the container using [Nomex Insulation Tape](#) or perhaps test using Kapton Tape.

The obvious downside to using Nomex Insulation is the fact that it is inherently less durable than FR-4 and will be prone to tearing. Fortunately, if it is taped in place, it will be easy to replace. It will be the responsibility of whoever is working on the Accumulator to check the state of the Insulation prior to reassembly. The author heavily suggests that an Accumulator Checklist is created. The purpose of this checklist is to ensure that critical elements of the Accumulator are checked every time there is maintenance or decommissioning.

EV.5.2.2 states that if the Container is made from a conductive material that the Segments and or Cells must be isolated from the inner walls of the container with an insulating

material. To ensure safety and compliance, the author suggests lining every surface of the Container with Nomex sheet. This is a decision made with safety in mind, with the volatile nature of LiPo cells and extremely large hazard of a short or electrical fire the engineers have a responsibility to keep the driver and team safe. It is suggested that the Lid, every Internal wall, Side Plate, and the entire Segment Cover should be fully insulated.

8.3. High Current Path

The High Current Path is shown in Figure 39.

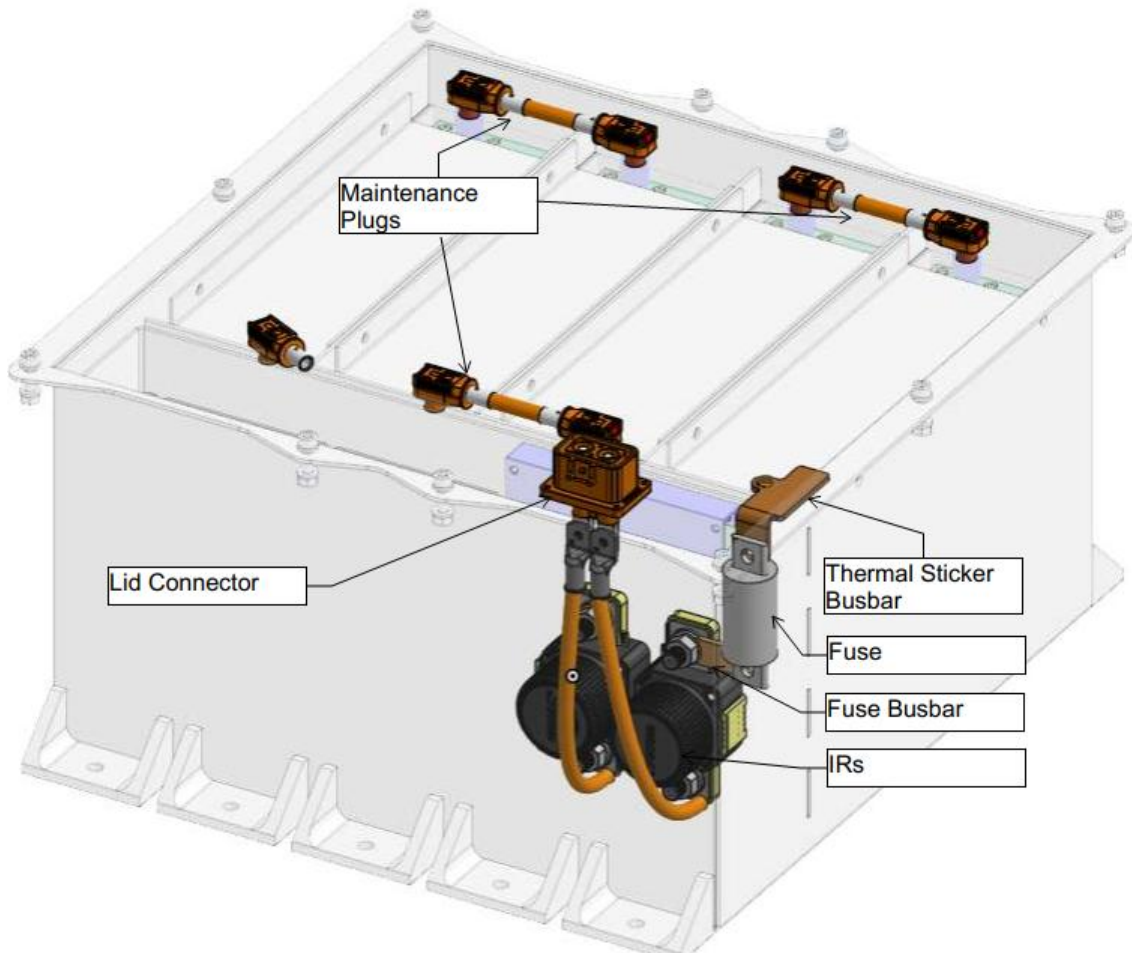


Figure 39. High Current Path (Excluding Segment Busbars)

The High Current Path (excluding the busbars in the Segments) includes the Maintenance Plugs, Lid Connector, IRs, Fuse, Thermal Sticker Busbar, and the Fuse Busbar.

Maintenance Plugs (required by rule EV.5.3) are plugs that must be able to be removed without tools once the Accumulator is open to allow for the separation of the Segments to reduce the pack voltage during maintenance and scrutineering. In the past Embedded Bud Connectors (EBC) also known as Methode Buds have been pressed into busbars to create one large Maintenance Plug known as the Service Handle, which is shown in Figure 40.

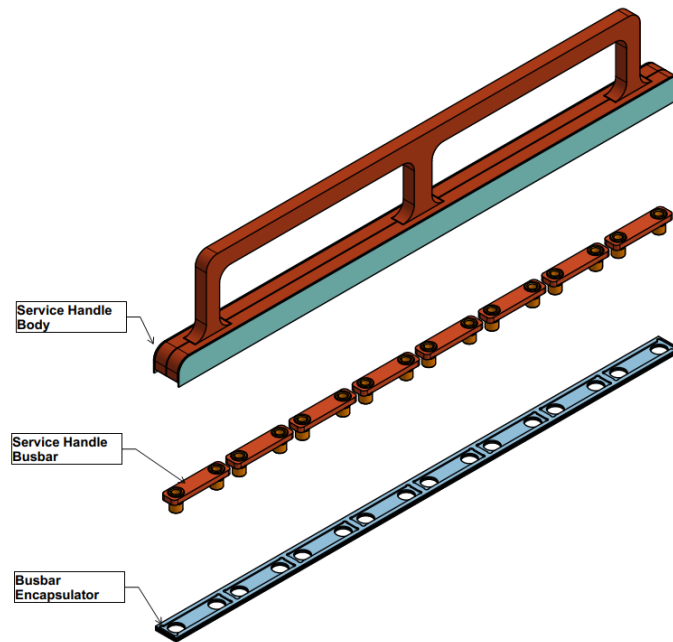


Figure 40. NU 24 Service Handle

The Service Handle of the past has been removed and instead replaced with three small HV wires with 5.7 mm Radlok Connectors attached on each side. The three Maintenance Plugs are the same length and can be interchanged.

Rule EV5.3.2 c. states that it cannot be possible to incorrectly install the Maintenance Plugs in an unintended configuration. Due to the length of the Maintenance Plugs it is physically not possible to connect them to the wrong pole of a different Segment.

EBC Connectors have been reused in the Thermal Sticker Busbar to fulfill rule EV7.5.8 in the FSAE-A Addendum which requires that teams have a place to mount a thermal sensitive sticker which is less than 30 mm away from the negative terminal. The Busbar uses one EBC connector that is placed onto a Methode Pin, which is the negative terminal of the Accumulator. This is shown in Figure 41. The distance from where the sticker will sit on the Busbar to the negative terminal is 7.3 mm.

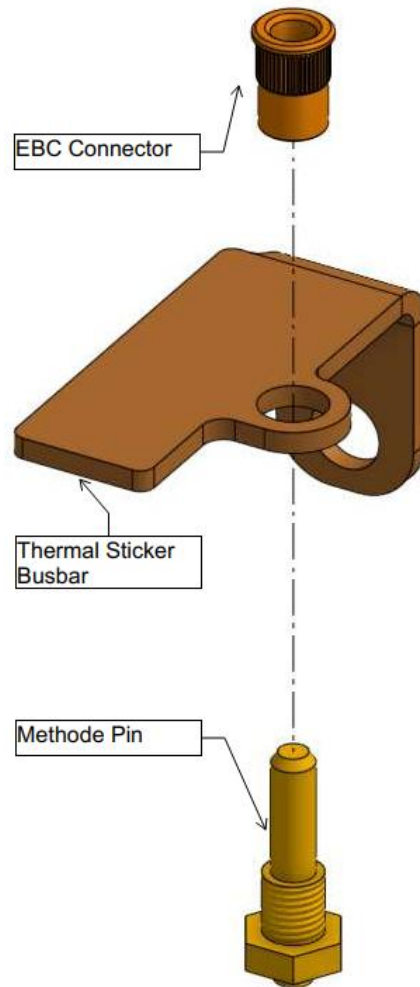


Figure 41. Thermal Sticker Busbar, EBC Connector, Methode Pin

It must be noted for manufacture that the dimension of the hole that the EBC Connector is press fit into that the hole is 9.235 mm in diameter. This is stated in the manufacturers technical drawing which is shown in Figure 42 alongside an EBC Connector.

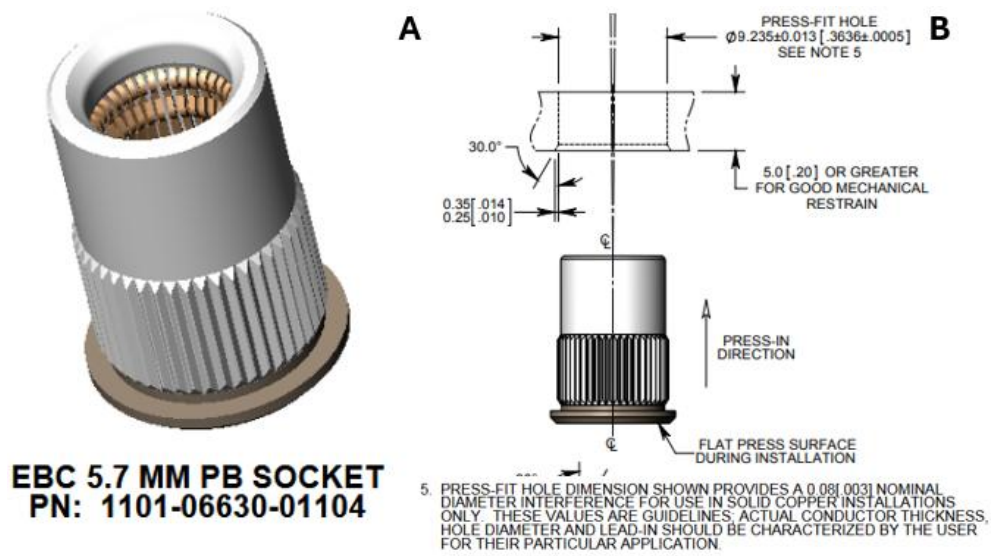


Figure 42. EBC Connector (A). Press-Fit Hole Tolerance for an EBC (B) [5]

The connector on the Lid has changed from the Surlok Connectors used in NU 24 to the ePower-Lite Connector from Amphenol used by the 2025 team (Part Number: ELPA2A2510).

9. CAD

As a result of the intention of this project to eventually be handed over, the Computer Aided Design (CAD) of the project was completed very deliberately.

Pending any large-scale changes that the author did not foresee, the CAD was intended to easily allow modification of values to suit measured real-life values. Examples of this include the tolerance of the Segment Housings to the Internal Walls of the Container and the thickness of the glue applied to the cells during their attachment to the Cell Holders.

Variable Studios were used extensively in the design to make it simple and easy to find and change different critical dimensions. An example of this is shown in the Container variables in Figure 43.

CONTAINER VARIABLES				...
Name	Variable type	Value	Description	
ACC_BASE	Length	4.06 mm	Accumulator Floor Thickness (FSAE Rules V1.0 2025) - (Thickness minimum from Calm Aluminium)	
ACC_WALLS	Length	2.54 mm	Accumulator Walls and Lid Thickness (FSAE Rules V1.0 2025) - (Thickness minimum from Calm Aluminium)	
WELD_LENGTH	Length	30 mm	Weld Length and Spacing Ratio for the Container (F.10.2.5.a FSAE Rules V1.0 2025)	
SLOT_TOL	Length	0.2 mm	Tolerance of the Internal Wall Slots (Welding Slots)	
SEG_BOLT_THK	Length	5 mm	The thickness of the Head of the Segment Housing - Cell Holder Bolts	
SEG_WIDTH	Length	103.4 mm	Width of a segment (for the Internal Walls)	
NOMEX_THK	Length	0.25 mm	Thickness of the Nomex Sheet	
SEG_CLR_TOL	Length	0.2 mm	Fitment Tolerance for the Segment (No Nomex)	
TOTAL_INTERNAL_...	Length	0.45 mm	Fitment Tolerance for a Segment (Nomex Thickness + Clearance Tolerance)	
HEIGHT_INT_WALL	Length	20.25 mm	Height of the Internal Walls over the Segment Housing Height	
Name				

Figure 43. Container Variable Studio

Updating these values, then updating context where required, mean that critical dimensions of designs can be easily altered.

A [video](#) showing how to complete some of these changes will be recorded and shared with the succeeding engineers.

10. Structural Equivalency Spreadsheet (SES)

The author's Final Design of the Accumulator passes the 2025 SES. To aid the understanding of the succeeding engineers, a video will be made detailing every input into the SES. This will be made available [here](#).

11. Calculations and FEA

To ensure some validity of the designs created, calculations and FEA, alongside the SES, were completed. These all concern the Segment Housings and its attachment.

All calculations were completed in an Excel document titled the [Accumulator Master Document](#).

It is suggested that prior to manufacture these calculations are repeated to ensure that no mistakes were made and that the system will behave as expected.

There are calculations and FEA that were not completed, or not completed to a high standard, these include:

- FEA of a Segment in any of the acceleration load cases
- Hand calculations of the for the loading of the Segment Housing

11.1. Segment Housing

The Segment Housing, as dictated by the rules (EV.10.3.1), must be able to withstand the following accelerations loads without the structure of the Accumulator Container and or the cells:

- 40 g in the longitudinal direction (forward/backwards)
- 40 g in the lateral direction (left/right)
- 20 g in the vertical direction (up/down)

The direction of these acceleration loads is shown in Figure 44.

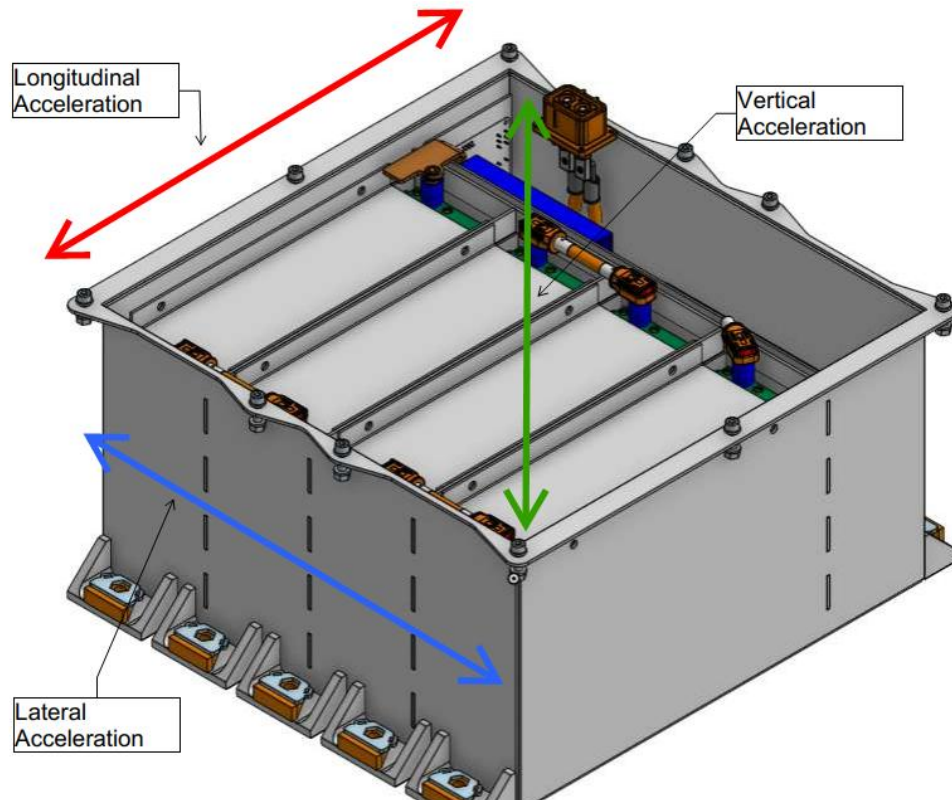


Figure 44. Direction of Acceleration Loads

11.1.1. Longitudinal Acceleration

With the current design, the direction of concern was the longitudinal force. The reason for this was that it placed the Cell Holders in bending. This loading is shown in Figure 45.

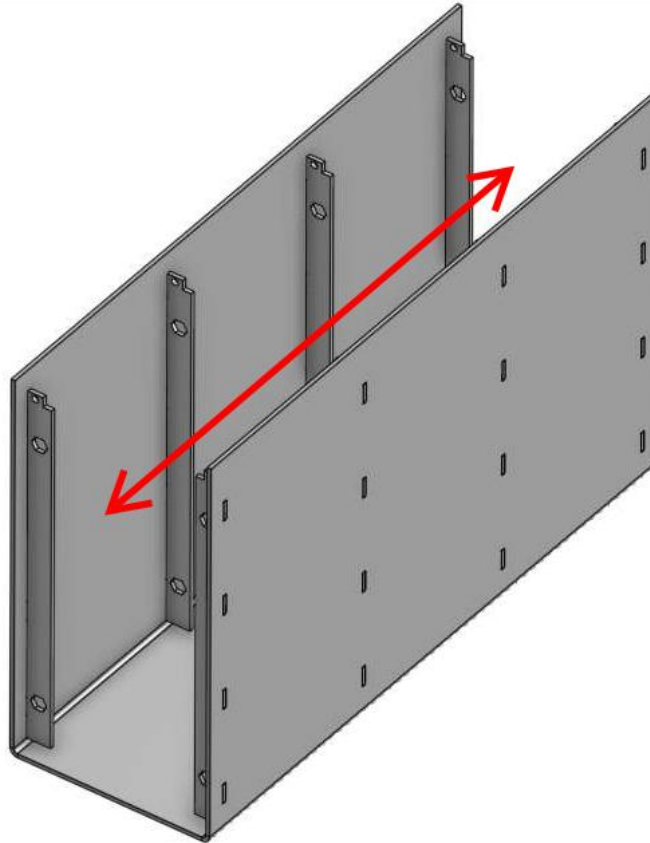


Figure 45. Longitudinal Acceleration Load

The acceleration loads are driven from the weight of the Segment. As no Segments have been manufactured a measured weight cannot be used. Utilising Onshape's weight measurement capability the weight of a complete Segment was estimated as 10.3 kg.

As this is only the structure of the Segment Housing, not including the Container or the cells, it is important that the boundary conditions of the solution are thought out before any calculations are taken.

The Segments are symmetrical, and therefore all calculations can be halved and completed on half of the Segment Housing.

Under the acceleration load the force generated by the cells only acts on the side of the Segment experiencing compression.

The compressive force would be pushed from each Cell Holder to each successive Spacer until it reached the last Spacer. From here the Load would then be applied to the final Cell Holder Mount.

A free body diagram (FBD) can be created of this scenario. This is shown in Figure 46

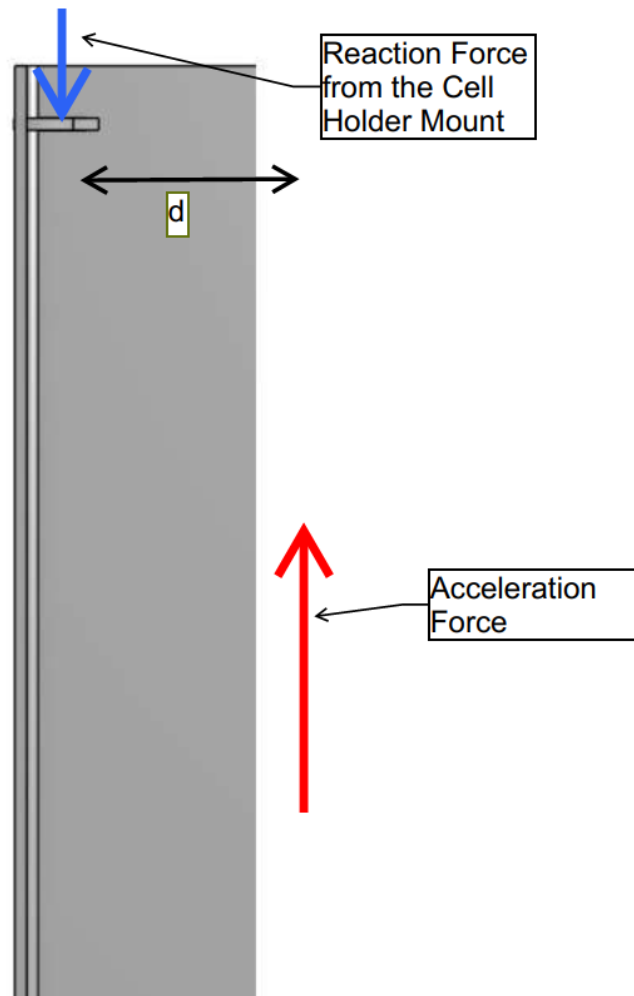


Figure 46. FBD of the Segment Housing (Longitudinal Acceleration)

From this FBD it can be shown that the Cell Holder Mount must react to a normal force from the Spacer, and a shear force at boundary with the Segment Housing. From this, calculations taking at the approximate height of a Spacer.

As the Cell Holder Mount is symmetrical about the cells, the acceleration load can be halved and the area around one spacer (one hex hole) in a Cell Holder Mount can be analysed.

The two forces form a plane stress, meaning that the stresses being analysed only acted on one plane.

The hand calculations for the stresses in the Cell Holder Mount are in the [Accumulator Master Document](#). Due to the size of the excel sheet it is not included in this report.

The calculation results are shown in Table 5.

Table 5. Longitudinal Acceleration Hand Calculations

Name	Magnitude	Units
Total Force	4041.72	N
Force due to Symmetry (on one side)	2020.86	N
Force distributed into cell groups (3 groups)	673.62	N
Assuming force placed on just the Compression Side (and evenly distributed on both Upper and Lower Spacers)	336.81	N
Spacer Area	34.13	mm ²
Normal Stress - σ_x	9.87	MPa
Direct Shear	6.13	MPa
A	329.40	mm ²
Von Mises Stress (Plane Stress)	14.50	MPa

Due to these results, it was unlikely that the Cell Holder Mounts would fail, and thus no FEA was carried out on these parts. It is still suggested that prior to manufacture this is completed.

FEA was completed on the Cell Holders themselves as these were likely to deform at their low thickness in the Longitudinal acceleration case.

The setup for the FEA is shown in Figure 47.

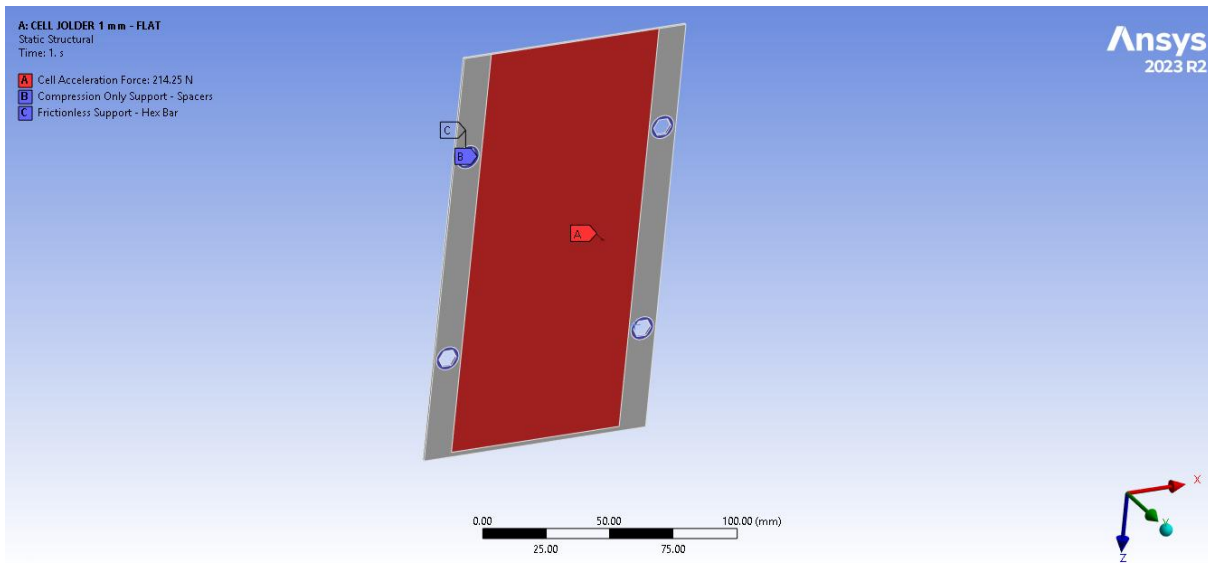


Figure 47. FEA Setup for the Cell Holders

A compression only support was used to mimic the 3D printed Spacer and a frictionless support mimicked the Hex Bar.

The Von Mises stresses are shown in Figure 48.

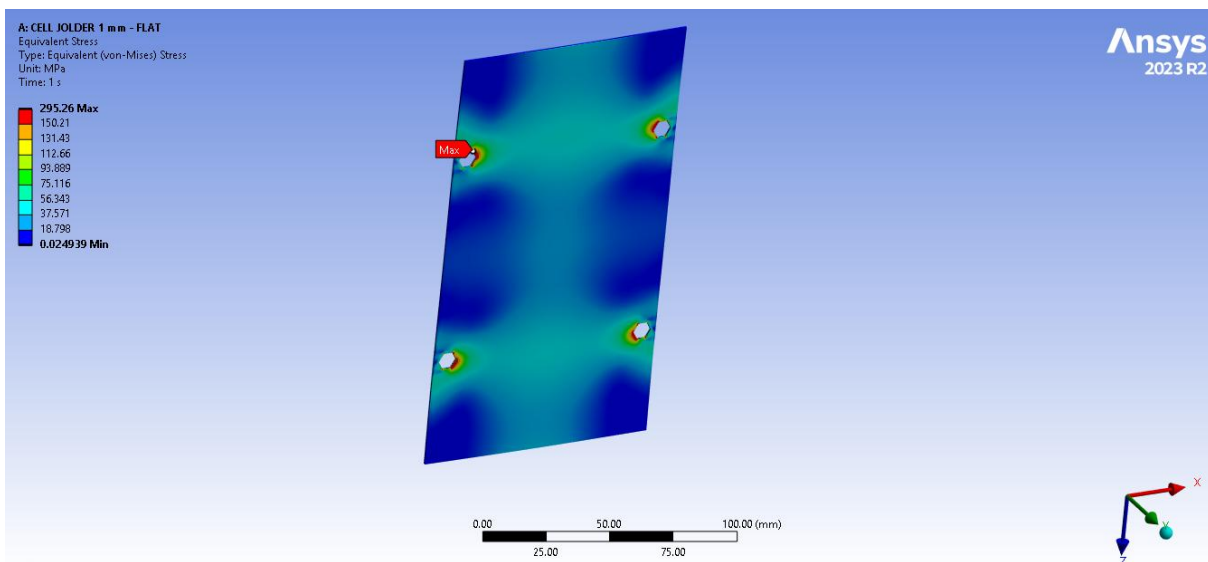


Figure 48. Von Mises Stress Distribution on the Cell Holder

The peak stresses surround the pointed edges of the hexagon cutouts for the Hex Bar. These are not stress concentrations limited to one node as shown in Figure 49.

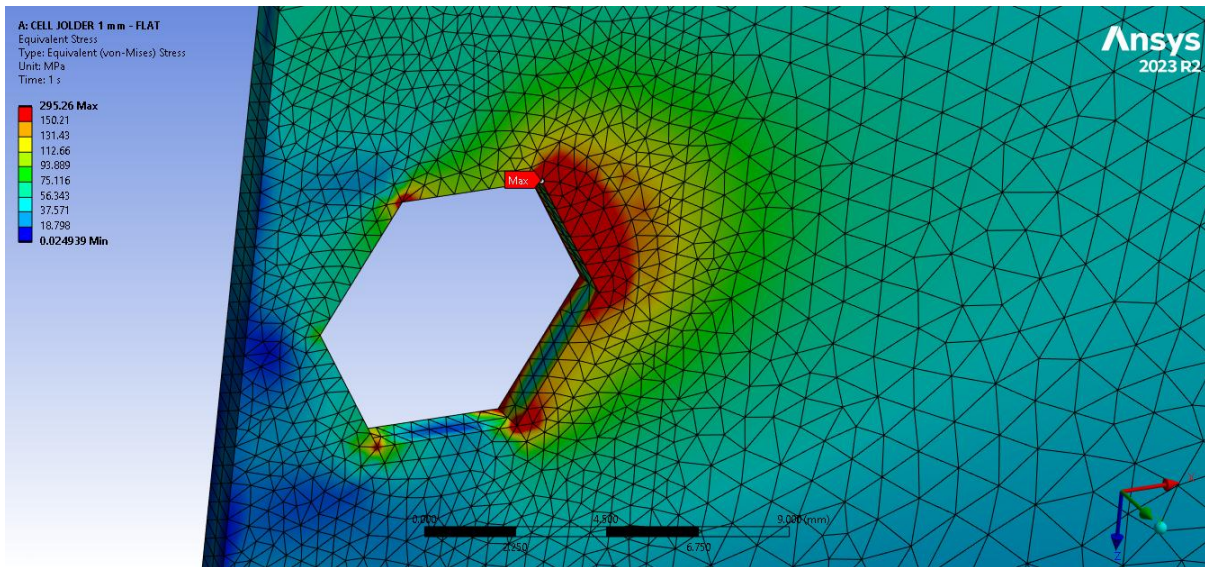


Figure 49. Stress Concentrations of the Cell Holder FEA

The stresses around the centre of the Cell Holder are shown in Figure 50.

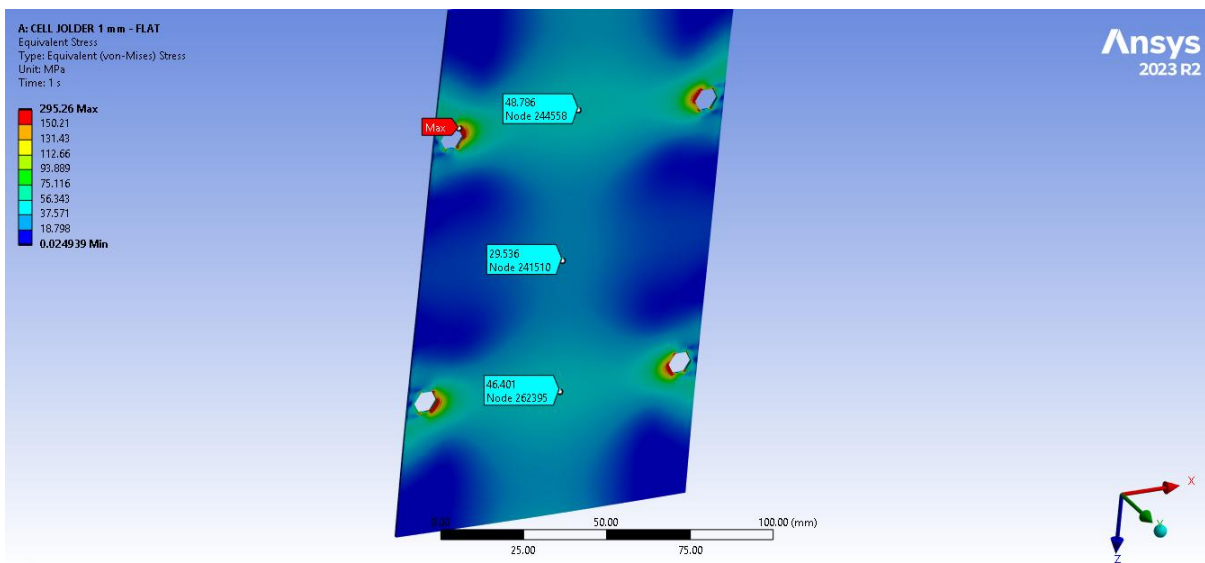


Figure 50. Probed Stresses in the Centre of the Cell Holder

The author suggests that the FEA is repeated without the Hex Bar cutout to see if the stress concentration reduces. If this does not occur, due to the cheapness and ease of waterjet cutting aluminium, it may be worth exploring the possibility of making a Cell Holder applying a load to it to see if it fails. The supplied load is only $2 \times \text{Weight of a Cell} \times 40$. This is approximately 21 kg. This being such a low weight means that testing (by simple means such as placing a

weight or getting someone to stand on it) is easy and not time consuming. If the Cell Holder does fail, the thickness of material can be increased.

11.1.2. Vertical and Lateral Acceleration

In the Vertical and Lateral acceleration loads, the Cell Holder Hex Bar is loaded equally by Cell Holders. This creates a large bending moment on the Cell Holder Hex Bar as shown in Figure 51.

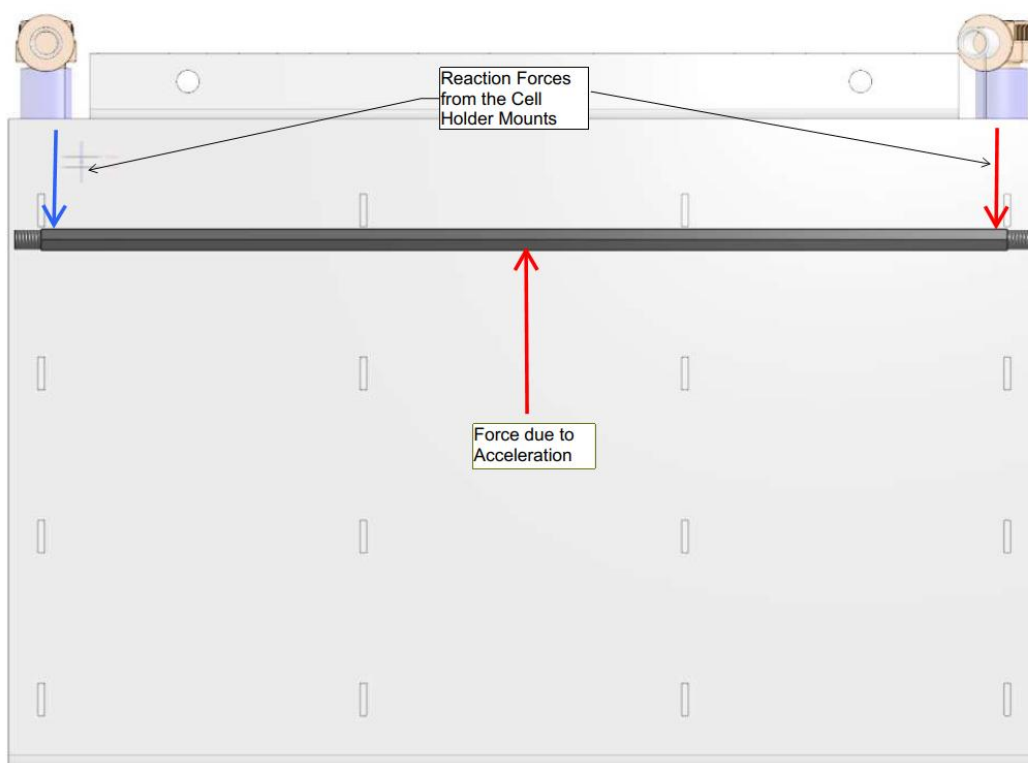


Figure 51. Bending Moment placed on Cell Holder Hex Bar

To reduce this large bending moment, four Cell Holder Mounts were implemented into the Segment Housing design. This would group the cells into three groups as shown in Figure 52.

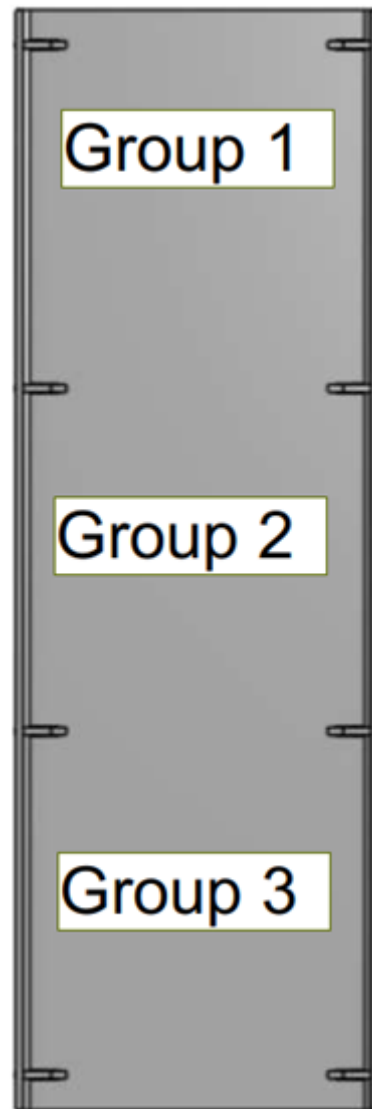


Figure 52. Cell Groups in a Segment Housing

Because the cells were now grouped, it meant the maximum force felt by the Cell Holder Hex Bar was now a third of the total load (in one place). This is shown by Figure 53.

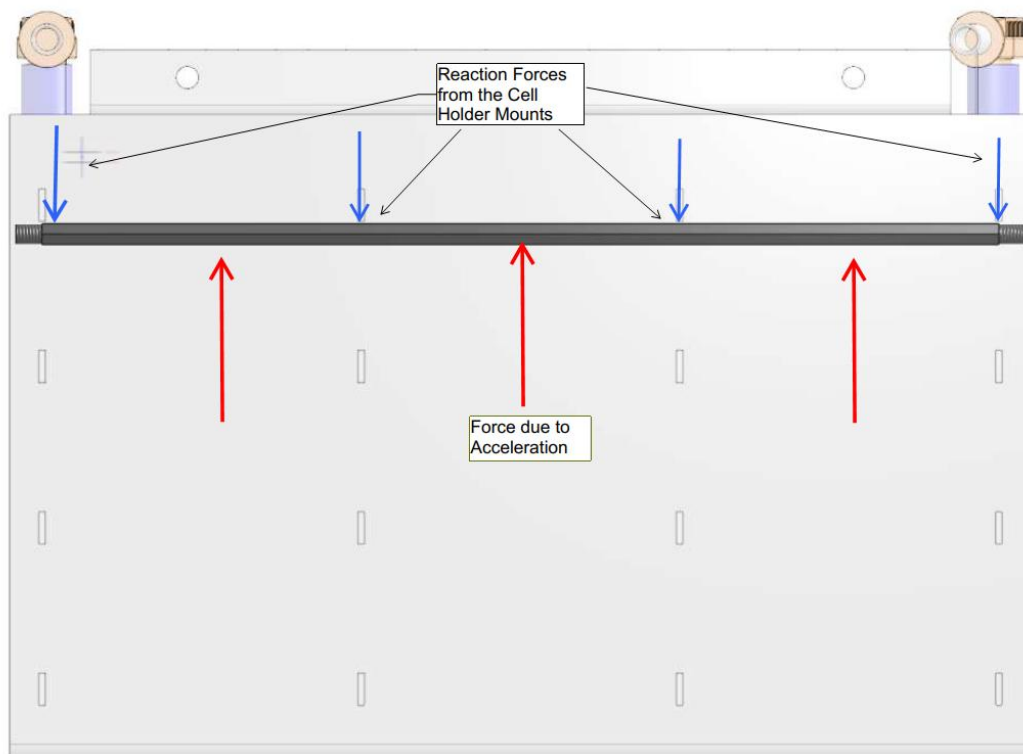


Figure 53. Bending Moment placed on Cell Holder Hex Bar with Four Cell Holder Mounts

As the Cell Holder Hex Bar was now supported in four positions, the maximum moment arm was decreased to only 48.7 mm.

The following assumptions were made in the rest of the calculation:

- The load applied to the Cell Holder Hex bar was applied in the middle of each cell group
- The applied force was the acceleration load of the cells and cell holders only
- All four Cell Holder Hex Bar perfectly shared the load
 - This meant that one Hex Bar could be analysed at 25 % of the total load

The results of the 40 g lateral acceleration, as this is the worst loading case for the Cell Holder Hex Bar, are shown in Table 6.

Table 6. Cell Holder Hex Bar Lateral Acceleration Calculations

Name	Magnitude	Units
Weight of just the cells and cell holder	8.07	kg
Force at 40 g	3168.35	N
Hex Area	34.92	mm ²
Hex Shear	22.68	MPa
Bending of the Hex Bar		
Max Bending occurs at the centre each beam section		
Beam Length	48.67	mm
Due to symmetry each beam carries the same load		
F (Total / Number of beam sections))	264.03	N
Moment	12.85	Nm
d (Across flat thickness of hex)	6.35	mm
I	97.72	mm ⁴
y (distance to centroid)	3.18	mm
Bending Stress	417.50	MPa
Von Misses Plane Stress	419.35	MPa

The Hex Bar designated for the is 12L14 mild steel which has a minimum to maximum yield strength range of 350-590 MPa [9], it is suggested that a length of material is purchased, then tested using a 3 point bending machine to observe the approximate yield and ultimate strength of the material. If it fails prior to the required loading, the size of the hex bar can be increased. But it should be noted that this is at a large weight penalty. The number of Cell Holder mounts can also be increased.

To ensure that the Cell Holder Mount would not fail from tear out another hand calculation was made. The assumption for this calculation was that the maximum reaction force generated at a Cell Holder Mount to be the full load of one group of cells accelerating at 40 g.

The results are shown in Table 7.

Table 7. Tear Out Calculation of Cell Holder Mount

Name	Magnitude	Units
Total Force @ 40 g	3168.35	N
Force from one cell group ((force / No. of hex bar) / no. of cell groups)	264.03	N
Tear out Stress (Shear)	18.00	MPa
Area (point to point length of hex * thickness)	14.7	mm ²

These results highlighted that it was extremely unlikely that the Cell Holder Mounts would fail due to shear tear out stress.

12. Prototype

As the intention of the 4200 project was for the author not to manufacture any components of the Accumulator, validation of the design would be difficult. The only form of physical validation, outside of calculations, was to make a prototype of the Container, Segment, and Side Plate.

Due to the intention of the CAD design to support changing variables, changing the thickness of the Container walls was quite easy. 3 mm thick MDF was selected as the material for the prototype Container. The dxfs for Container walls and floor were exported from Onshape and the MDF was cut in the Architecture and Built Environment Workshop (ABEW). The assembled Container is shown in Figure 54.



Figure 54. Assembled MDF Prototype Container

To create a prototype of the Side Plate the geometry was simplified so it could be laser cut from a single piece of the same 3 mm MDF. The two flanges of the Side Plate were removed, and the Side Plate was laser cut with the rest of the Container. Following this, due to the tight

tolerances between IMD, Fuse, and BMS, footprints of the IMD and the BMS were 3D printed as well as the Fuse Busbar and the Thermal Sticker Busbar. In an ideal world the real IMD and BMS would have been used however, during this period removing these parts from the Accumulator was not possible during track days. These parts were fully assembled and are shown in Figure 55.

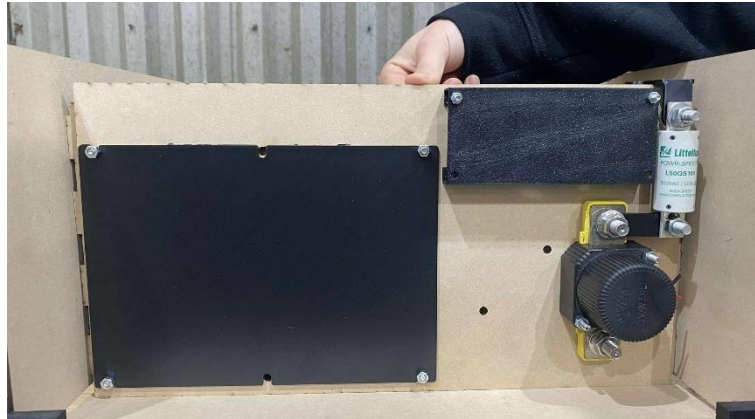


Figure 55. Assembled Prototype Side Plate

In the CAD the minimum distance between the IMD and the HV Fuse was 3.7 mm and the IMD Cover designed was interfering with the Fuse. Once the prototype Side Plate was assembled, it was quite clear that there was a lot more space than in the Onshape assembly. The author checked that the HV Fuse was the correct fuse, and once learning this realised that the Fuse in the CAD was over 5 mm larger in diameter than the Fuse was in real life, meaning that the Fuse and IMD/IMD Cover did not have clearance issues and would be fine in their current positions.

The initial concern about the clearance between the IMD connector and the BMS were also alleviated. However, as only a footprint of the BMS was 3D printed, it cannot be certain that the clearance is fine. Prior to manufacture the real BMS **must** be attached to the Prototype Side Plate, alongside the real IMD with all their connections plugged to ensure proper clearance.

The Segment were also prototyped. The Segment Housing and Cell Holder Mounts were 3D printed as one part split into two halves to fit on the print bed of the Bambu P1S printers available. A similar process was undertaken for the cells and Cell Holders, where one part that represented two cells and one Cell Holder was created and printed. The Upper and Lower Busbars, the Busbar Holder, Cell Holder Hex Bar, and spacers were all also 3D printed. With all the associated parts, they were assembled with relative ease and placed into the laser cut MDF Container. This is shown in Figure 56.



Figure 56. Prototype Segment and Side Plate in MDF Container

During design one of the main concerns of the assembly of the Segments was placing the Spacers over the Cell Holder Hex Bar inside of the Segment Housing. To the authors surprise, this process was quite easy, with the width of the Segment Housing being large enough to see into and manoeuvre the parts into position.

The fitment tolerance between Container and the Segments was opened to 0.4 mm per side (0.8 mm total) from the current value of 0.2 mm per side. The effect of this was no recognised as the 3D printed Segment and MDF Container were both flimsy and would simply allow for

movement where flexing should not have been possible. This meant that the prototype did not allow for validation of the fitment of the Segments into the Container.

13. Next Steps

As this design is handed over, there are works that need to be completed prior to the commission of the NU 26 Accumulator. These are broken into three categories: Design, Physical Testing, and Manufacture.

13.1. Design

As highlighted throughout this report there are still design work and validation to be completed.

The remaining design work is:

- Design of the Segment mounting insulation (as discussed in Container)
- Detailed FEA and validation of the Segment Housing designs
- Design and commission of a Precharge Board and LVD board that will fit the Side Plate / Lid respectively
- Design of a Segment PCB
- Finalisation of a Segment once the Segment PCB is designed
- Design of an insulative cover that sits on top of the Segment on top of the Segment PCB

13.1.1. Rules Enquiry

The most pressing design change that may need to occur is the mounting of the Side Plate and the Segment Housing to the Internal and External Walls.

During the middle of May 2025, the author sent the following rules enquiry to FSAE-A:

“Rule F.10.3.2.b. states that segments can be held by "Mechanical Segment attachments to the container". In reference to this, is the segment housing allowed to be purely bolted to the internal walls of the container, provided it is above the insulative 'lid' of the segment and both the head and nut side of the fastener assembly are covered (the same as mounting hardware for

the container in EV.5.2.2) and there is an insulative material between the segment housing and the internal wall?

To extend on this if one bolt is used to bolt two segments (i.e. segment housing 1 - internal wall - segment housing 2) and it is covered does this comply to rule EV.5.2.3 (assuming it meets the acceleration requirements for mounting the segments)? Just reiterating that the bolt will be above the insulative 'lid' of the segment so it is insulated from the Tractive System.

This is seemingly done in FS Austria (https://fsaustria.at/wp-content/uploads/FS4A-Session3_Electrical-Design_and_Scruti.pdf, Page 13). If it is not done in this manner, is there another way that FSAE-A deems acceptable?

Images are attached for reference (please note there is not cover on the bolt/nut assembly, instead where the covers would be in just highlighted in red).

Thank you for your consideration.”

The attached image is shown in Figure 57.

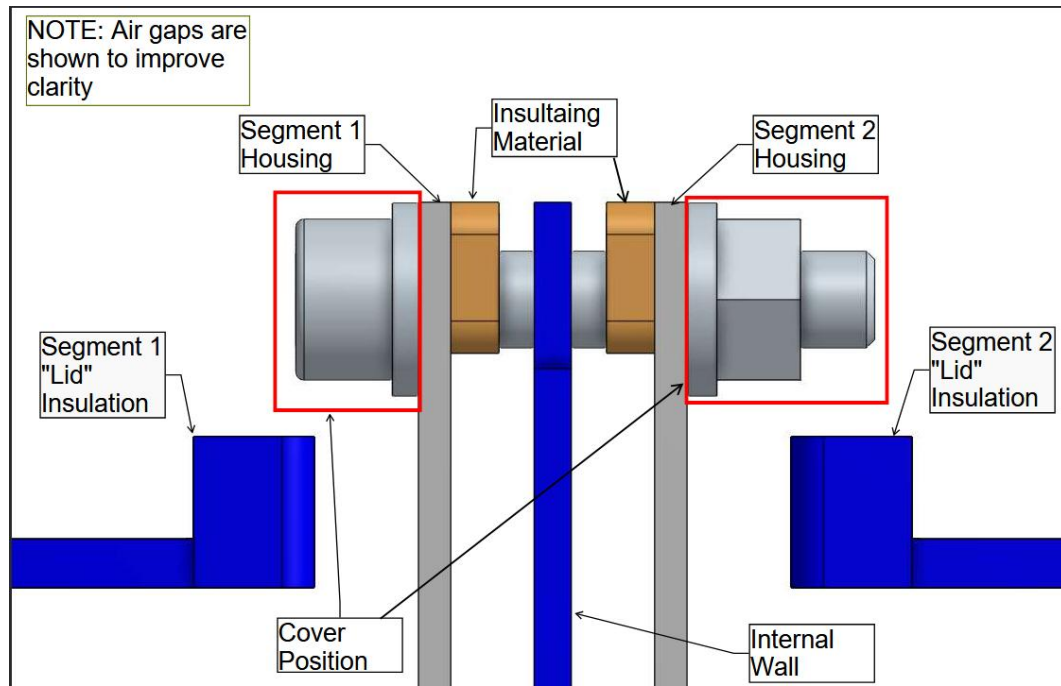


Figure 57. Segment Mounting Rules Enquiry

Prior to this rule's enquiry, the author had intended to bolt the Segment Housings directly to the Internal and External Walls, however, was unsure of its legality as the Housings were made of aluminium. Due to the incoming due date of this report, the author modelled the Segment Cover for holding the Segments in position. Pending the results of the enquiry the following action should be taken.

If this is allowed, remove the Segment Cover and bolt the Segments to the Internal and External walls. This will save weight by removing the need for the Segment Cover and will also reduce part count.

If it is not allowed, keep the Segment Cover or remove it entirely for a new design. If the Segment Cover is kept, it must have corresponding calculations and FEA completed on its design to ensure that it meets rule F.10.3.2 b. It must also be completely wrapped in Nomex Paper (to reduce risk of shorting if a failure occurs within the Segment).

13.2. Physical Testing

Physical testing will take time and creation of thoughtful testing procedures. What testing is suggested (at a minimum) is:

- Testing of the lap shear strength of the 834B Epoxy for the Cell – Cell Holder bonds
- Testing of the Cell Holders strength – especially in the longitudinal direction
- Testing the thickness of the Cell Holder – Cell bonds
 - This must be done such that the Segment Housings and Container can be manufactured to be the correct size
- Testing on the cooling characteristics of the new cells and Accumulator (post manufacture)
- Testing on all the electrical components (post manufacture)

13.3. Manufacturing

Once manufacturing is ready to commence, it is recommended that initially one Segment is made only. This will allow for the thorough testing and validation of the gluing method to secure the cells but also allow for measurements to be taken of the Segment length and width. This will allow for the Container dimensions to be finalised, notably the distance between the Internal Walls. It will also give the chance to test assembling a full Segment for the first time and fully insulating it. This ‘test’ run must occur before full manufacturing takes place.

14. Handover

The handover process for this design is required to take considerable care. The reasoning for this is that it is a design that will be undergo handover at least twice. From the author to the 2025 team, then from them to the 2026 team.

The large issue with this is the risk of loss of information. To attempt to try and hinder this, this report has attempted to highlight a lot of the design decisions, as well as scope, goals, and constraints and how these affected the design process.

In addition to this report two more resources are being created to assist the handover. A presentation will be delivered by the author to the associated 2025 NU Racing engineers. The 2025 engineers are expected to make large headway with the physical testing and manufacturing of the Accumulator.

As they will not be the team that uses this Accumulator design, a video presenting the same information will be created by the author, with the 2026 team as the intended audience.

The main hope is that a repeat of 2024 is avoided. In that year the team inherited designs from the previous team and many members of the team did not understand the design decisions that went into creating NU 24.

There will be three videos available. One that summarises this report, another will summaries how to update the CAD model using Variable Studio and Modelling in Context, and the final video will explain the entries into the SES. These will all be available in the [NU Teams SharePoint](#).

14.1. Presentation

14.1.1. Q&A

It is not expected that after reading this report and viewing a presentation that the 2025 team will fully understand every thought that the author had during the design process. And due to

this the author has organised that at minimum one Q&A session will take place. The intention is that one session will take place immediately after the presentation, and one session will take place in the days or weeks after the presentation. This is when the audience will have had time to digest and comprehend the available information. Hopefully after giving the audience time to review the presentation, this report, and any other materials that are of interest, more in depth and meaningful questions can be asked. The author believes that this is incredibly important to allow for as much knowledge transfer as possible.

14.2. Video Recording

14.2.1. Building from Presentation and Q&A

The Video Recording, especially for the video that will summarise this report, will need to build from the Presentation and corresponding Q&A sessions. This is because the author may not be able to deliver the presentation again to the 2026 team.

Due to this it is incredibly important that the recorded videos can be as effective as possible. A great way to learn what is effective / what needs to be improved for the video is the presentation itself.

14.3. Transition Support and Communication

No transition period is seamless, and the author anticipates this. During the period before the Accumulator is commissioned the author is happy to be contacted with any questions or concerns. This design is still theoretical, and there is every possibility that when it is manufactured there may be issues.

15. Reflection

15.1. Assessment Against Scope and Goals

The scope for the project was as follows:

- Design of:
 - Container
 - Segment Packaging
 - Top Plate
 - Mounting
 - Lid
 - Maintenance Plugs
- Completion of the Structural Equivalency Spreadsheet (SES)
- Handover of the project design at its completion

Every task was complete (except for the handover at the time of writing).

The goals for the project were:

- Have 10 kg minimum in weight reduction
- Reduce footprint of the Accumulator
- Maintain serviceability

Due to the nature of this project, none of these goals can be defined as successes. Without the final working system, these goals are still in limbo but are trending in the correct direction.

At the time of writing the approximate weight and volume reductions from the NU 24 Accumulator to the NU 26 Accumulator are shown in Table 8

Table 8. Weight and Volume Comparison of NU 24 and NU 26's Accumulators

Variable	NU 24	NU 26	Difference
Accumulator Weight [kg]	70	53	-17
Volume [L]	75	51	-24

It must be noted that the NU 24's Accumulator weight was physically measured so it included the mass of the wiring, connectors, and other miscellaneous parts. Whereas the weight

measurement of NU 26's Accumulator only includes what is in the CAD model. The difference when the Accumulator is manufactured will be less than the 17 kg it is currently.

15.2. Impact of the Design

If this Accumulator can be commissioned and remain saving approximately 15 kg in weight and 20 L in volume is massive to the scope of NU 26. This will mean that the total car mass could improve all the way down to 225 kg (NU 24 was 242.5 kg) from one system redesign. This will undoubtedly see great performance increases and places NU Racing in a position to make their car one of the lightest in FSAE-A.

The reduction in volume implies a few things. The Chassis and surrounding systems (like the Tractive System Stack) may be able to decrease in width and length further increasing weight gains.

As this is a first iteration design, there are undoubtedly improvements that will be able to be made in the coming years. Looking at the NU 24 Accumulator shows that over several years of different engineers and mindsets, the system can continue to improve drastically. With optimism it is hoped that the same can be achieved with this design.

15.1. Lessons Learnt

At the completion of this project there were many lessons learnt. Some which are shown in

Table 9.

Table 9. Lessons Learnt Table

Issue / Success	Description	Recommendations (If any)
Issue	The difficulty in taking on a technical and workload heavy project following a leadership role	Manage workload and ease into a large project. Take time for yourself
Issue	With such a large design load, many of the calculations and FEA validations were left too late to be useful, fully completed, and/or included in this report. This has greatly reduced the impact of the project	Do not leave important results later than required and attempt to rescope or redistribute resources to assist hitting milestones
Success	The author was unfamiliar and at times out of depth with aspects of an electrical design	Ask for the opinion experience people who have done a similar task before. The author relied heavily on the Chief Engineer, Alec Chapman, and Student Teams Managing Engineer, Malcolm Sidney.
Success	Creating robust CAD models when it is known that a design may undergo mass changes	Think about critical dimensions of parts and models and try to model around them

16. Key Takeaways / Conclusion

The inherent flaw with this project was that very little engineering validation could occur. This meant that at the end of the project, the author would have no real-world deliverables that were measurable and meaningful. This was known from the beginning of the project.

What is now available to the 2025 and 2026 team is a platform to build from and hopefully improve / change the design that is currently laid out for them.

The importance of knowledge transfer, and whether NU Racing has learnt from its previous mistakes, is ever present in this project and will only be able to be measured at the completion of the 2026 competition. It is with great hope and optimism that the author looks forward to seeing this Accumulator come to life.

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Appendix A

Table 10. Vertical Friction Force Required for a Compression Based Design

Vertical Force			
		Units	Notes
g	9.81	m/s ²	
	20	-	
w	7.371	kg	Weight of the segment (just cells)
mu	0.8	-	Approximate friction coefficient
Vf	1446.1902	N	Vertical Force = $w \cdot g \cdot 20$
The minimum friction force (Ff) must be greater than the vertical force			
Ff	1807.73775	N	Minimum friction force required = Vf / μ
Ff	184.275	kg	Ff in kgs
P	2714.320946	kPa	Pressure over the cell